

Applications of Artificial Intelligence in Music: A Review

Wen Ma¹ and Zhong Zheng¹ ✉

ABSTRACT

Crowd science is a science that uses the collective intelligence of information, physics, and society as a whole in the context of large-scale online interconnection to improve social and economic efficiency. Artificial Intelligence (AI) has become an increasingly influential force in the field of music, reshaping traditional paradigms of creativity, production, performance, and consumption. This review examines the current applications of AI across key musical domains, including composition, audio production, performance, analysis, recommendation, and interdisciplinary integration. The paper highlights how AI techniques—ranging from symbolic rule-based systems to advanced deep learning architectures—have enabled novel capabilities such as automated melody and lyric generation, live collaboration between humans and AI agents in musical performance, sentiment-aware music analysis, and personalized recommendation systems. Additionally, the review explores the transformative effects of AI on the structure and ecology of the music industry, as well as its expanding role in education and therapy. Despite these advances, unresolved challenges remain in areas such as interpretability, ethical accountability, and the definition of creativity. The paper concludes by outlining future research directions, emphasizing the importance of human-AI collaboration, standardization of methodologies, and the development of adaptive AI systems for live musical contexts. By synthesizing recent progress and identifying open questions, this review provides a foundation for understanding the evolving relationship between AI and music.

KEYWORDS

artificial intelligence; music recommendation; music consumption; music industry; crowd intelligence

In recent years, the integration of human and artificial intelligence has become a key driver of innovation across various domains. Crowd science is a science that uses the collective intelligence of information, physics, and society as a whole in the context of large-scale online interconnection to improve social and economic efficiency^[1]. The field of music is a vivid and practical embodiment of this principle. In music creation, distribution, recommendation, and consumption, Artificial Intelligence (AI) systems do not operate in isolation but in collaboration with human composers, performers, producers, and listeners. This human-AI collaboration exemplifies the core idea of collective intelligence: leveraging machine learning, neural networks, and algorithmic curation alongside human intuition and creativity. Whether in co-composing new musical works, optimizing music marketing through AI-based recommendation engines, or enhancing fan engagement with interactive AI-driven performances, these applications demonstrate how crowd science is being realized in practice. Thus, the evolving role of AI in music is not merely technological advancement but also a social experiment in augmented intelligence and collaborative creativity.

1 Introduction

AI refers to the development of computer systems capable of performing tasks that typically require human intelligence, such as reasoning, learning, perception, and decision-making^[2]. In recent years, rapid advancements in Machine Learning (ML), particularly deep learning, have enabled AI to achieve remarkable performance in fields such as image recognition, natural language processing, and autonomous systems^[3]. These breakthroughs have also sparked a transformative wave in the creative arts, especially

in the domain of music.

The application of AI in music has undergone a significant evolution. Early efforts, dating back to the 1950s, involved rule-based systems designed to generate simple compositions using pre-programmed music theory^[4]. These systems, while pioneering, lacked the ability to learn from data or adapt to stylistic variation. The emergence of ML introduced statistical approaches that enabled algorithms to analyze musical patterns from large datasets and generate compositions with greater complexity^[5]. The advent of deep neural networks, including Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) networks, and transformer-based models, has since revolutionized the landscape of AI-generated music, allowing for more coherent and expressive outputs^[6].

Today, AI technologies are applied across a wide range of musical processes—from melody and lyric generation to audio production, performance enhancement, sentiment analysis, and personalized recommendation systems. These applications not only augment the creative capacities of artists but also reshape how audiences interact with music^[7]. Moreover, AI is increasingly intersecting with interdisciplinary domains such as music education and therapy, raising new questions about creativity, authorship, and the ethics of machine-generated art^[8].

While several previous reviews have surveyed specific aspects of AI in music—such as composition techniques, generative models, or music information retrieval—this paper distinguishes itself by offering a comprehensive and integrative overview that spans the entire music ecosystem. It brings together technical innovations, practical applications, and emerging social and ethical considerations in a unified framework. Furthermore, this review emphasizes recent trends, such as the role of large-scale generative

¹ Faculty of Music, Macau University of Science and Technology, Macau 999078, China

Address correspondence to [Zhong Zheng, zzheng@must.edu.mo](mailto:zhong.zheng@must.edu.mo)

© The author(s) 2025. The articles published in this open access journal are distributed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>).

models, real-time interaction in live performances, AI-enabled music education, and therapeutic uses of AI in musical contexts—areas that have received limited coverage in earlier literature.

The primary objective of this review is to provide a comprehensive examination of current applications and emerging trends in AI-driven music systems. The paper aims to explore technological advancements in composition, production, performance, and analysis; the transformation of music consumption and recommendation; the broader structural impacts on the music industry; and the expanding frontiers of interdisciplinary research. By synthesizing contemporary literature and case studies, this review seeks to illuminate both the innovative potential and the complex challenges that AI presents in the evolving music ecosystem. Figure 1 is the main content of this article.

2 Technological Application of Artificial Intelligence in the Field of Music

2.1 Music composition

2.1.1 Melody generation, harmony, and arrangement

AI has significantly advanced melody generation, evolving from early statistical models to sophisticated deep learning techniques. Initially, Markov chains modeled melodies as stochastic processes, predicting subsequent notes based on preceding ones^[9]. While effective for short-term dependencies, they lacked the capacity to capture long-term musical structures. Grammar-based and rule-based approaches, inspired by linguistic theories like Chomsky's grammars, utilized predefined rules to generate melodies, offering stylistic consistency but limited adaptability^[10–12]. Evolutionary algorithms such as GenJam use genetic algorithms to generate music works, relying on manual feedback as a fitness function, which is less efficient but performs well in terms of exploration and diversity^[13].

The advent of deep learning brought transformative changes. RNN, particularly LSTM, enabled the modeling of temporal dependencies in music^[14]. Enhancements like Lookback RNN and Attention RNN improved the capture of musical repetition and focus^[15]. Models such as Deep Artificial Composer (DAC) and Song From PI employed hierarchical RNN to jointly learn pitch and duration, facilitating the generation of complex musical pieces^[16,17]. Some models allowed for the modeling of larger

musical structures, such as MusicVAE, Music Transformer, and StructureNet^[18–20]. These models can generate melodic fragments with diversity and coherence structured musical compositions. However, careful parameter tuning is required when using these models^[21].

With the advancement of AI technologies, traditional harmony, and arrangement approaches, which rely heavily on the rules of music theory, are being increasingly replaced by more intelligent generative methods. AI systems are capable of automatically generating music by leveraging deep neural networks trained on large datasets of harmonic structures, rhythmic patterns, and chord progressions. While earlier methods were primarily rule-based, recent approaches have shifted towards data-driven models centered around RNN, Convolutional Neural Networks (CNN) and Generative Adversarial Networks (GAN), moving progressively towards styles that are more diverse, structurally complex, and melodically natural.

Specifically, Coconet utilizes CNN to model multiple voice parts in Bach chorales, filling in incomplete musical scores by randomly masking notes, thereby demonstrating CNN's potential in modeling the spatial structure of music^[22]. DeepBach employs bidirectional LSTM to simultaneously incorporate past and future musical contexts, improving the coherence and musicality of generated melodies^[23]. Folk-RNN and Song From PI apply LSTM architectures to model folk music and layered musical structures, enabling multi-level control in generation^[24]. MuseGAN introduces a multi-generator GAN framework to achieve breakthroughs in multi-track arrangement, though its early versions suffered from overly fragmented note sequences. BinaryMuseGAN addresses this issue by incorporating binary neurons and semitone feature detection, improving note continuity^[25,26]. LeadSheetGAN utilizes lead sheets as conditional input, which enhances melodic directionality and thus yields musically expressive outputs^[27].

In terms of interactivity and personalization, MetaCompose uses multi-objective adaptive functions to generate melodies and chords, although it lacks robust harmony generation capabilities. EvoComposer compensates for this limitation by integrating classical music rules with statistical patterns extracted from musical corpora, enabling style imitation of specific composers^[28,29]. Additionally, Kaliakatsos applies particle swarm optimization and genetic algorithms to allow users to guide the generation process by specifying preferred rhythmic and pitch features, thus enhancing personalization^[30]. These approaches collectively demonstrate the immense potential of integrating deep

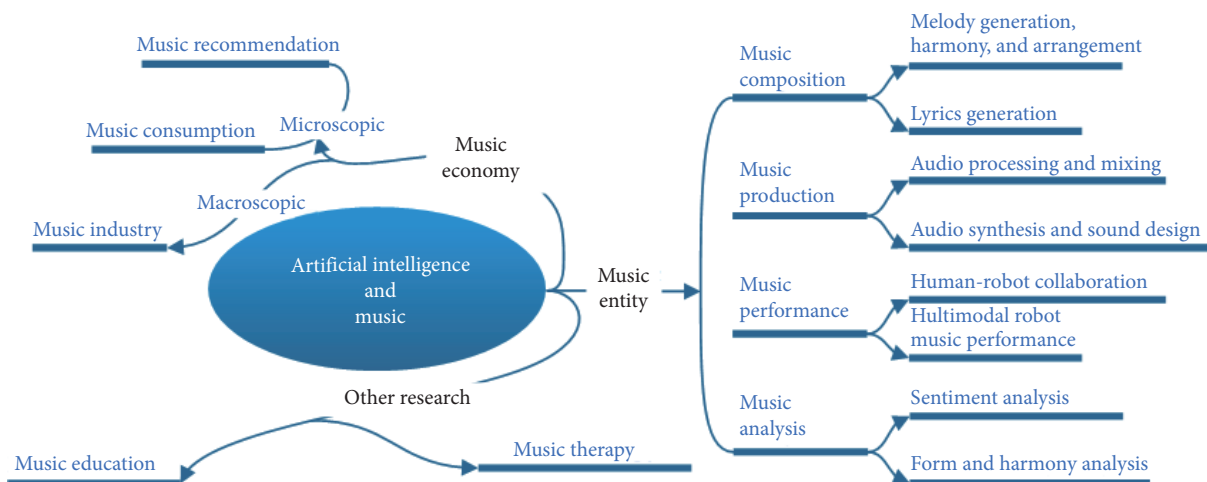


Fig. 1 Research contents.

learning with domain-specific musical knowledge, steering automatic harmony, and arrangement generation towards greater intelligence, adaptability, and stylistic diversity.

Recent advancements integrate Large Language Models (LLM) into symbolic music generation to enhance melody, harmony, and arrangement processes. ByteComposer emulates human compositional logic through a four-stage LLM-agent framework, enabling interpretable and interactive melody generation^[31]. The paper shows that by fine-tuning a pre-trained music language model for a multi-track arrangement task, it demonstrates superior performance and control in different instrument settings^[32]. SongComposer, the first unified LLM for simultaneous generation of lyrics and melody, leverages symbolic representation and progressive structure-aware training. Collectively, these models exemplify the transformative role of LLM in structured, human-aligned music composition. Table 1 summarizes and compares models from different periods.

2.1.2 Lyrics generation

Research on AI-generated lyrics has evolved significantly from early poetic models to more sophisticated lyric-writing systems. Initial lyric generation relied on probabilistic models such as Markov Chains and RNN, often combined with rhyme and meter enforcement techniques^[33–35]. Hip-hop and rap lyrics research introduced retrieval-based methods and Support Vector Machines (SVM) to maintain semantic coherence^[36], while projects like WASP and Rap Lyrics Generator experimented with rule-based and n -gram models, though often producing incoherent or theme-less lyrics. Recently, transformer-based models, SeqGAN, and domain-constrained language models have been applied to enhance lyric quality, control, and theme alignment^[37,38]. However, many systems still lack mechanisms to ensure singability or musical prosody alignment, which is crucial for melody compatibility^[39–41].

In parallel, Singing Voice Synthesis (SVS)^[42,43] and music generation have made strides through the use of transformer models, diffusion models, and Latent Diffusion Models (LDM)^[44,45], enabling high-fidelity, expressive musical outputs (MeLoDy^[46] and AudioLDM2^[47]). Tools like SongCreator^[48] integrate lyrics, melody, and accompaniment generation, marking a shift from traditional SVS that relied on human-composed scores. While early music generation models focused on symbolic output, newer systems tokenize music for text-based processing and blend LM^[49] with LDM for holistic audio synthesis. Despite these advances, most remain limited to either instrumental or non-lyrical outputs, with few integrating lyrics and vocals simultaneously.

Emerging work highlights a critical gap in the explainability

and prosody of lyric-generation systems. Few models address musical prosody or provide interpretability, limiting usability for composers and singers. Explainable AI (XAI) approaches in music computing have visualized latent spaces or generated human-like rationales for recommendations^[42,50,51], yet these explanations often lack musical specificity. Recent studies have explored the use of LLM in lyric generation, focusing on alignment, stylistic control, and metadata conditioning. A multi-agent GPT-based system has been developed for Mandarin lyrics, enabling rhyme control, tonal congruence, and syllabic precision^[52]. Another approach introduces a unified LLM for simultaneous generation of lyrics and melody using structure-aware training techniques^[53]. Additionally, reconstructed lyrics have been generated from Bag-of-Words data using LLM guided by genre, mood, and artist metadata, circumventing copyright concerns^[54]. These works showcase LLM's creative and contextual flexibility. Overall, future research must prioritize integrating musical prosody, semantic alignment, and explainability to produce singable, meaningful lyrics^[55,56].

2.2 Music production

2.2.1 Audio processing and mixing

AI-powered mastering platforms have democratized access to professional-grade audio optimization. Tools like eMastered and LANDR leverage ML to analyze audio tracks, applying genre-specific algorithms to adjust frequency balance, dynamic range, and loudness^[57,58]. For intermediate producers, these systems offer time-efficient solutions: eMastered's AI can process tracks in minutes, providing multiple mastered versions tailored to industry standards, thereby eliminating the need for costly human engineers or specialized equipment^[57]. LANDR's supervised ML models learn from vast datasets of professionally mastered music, emulating human decision-making to standardize sound quality^[59]. However, limitations arise in handling artistic diversity: AI struggles to maintain cohesive tonal balance in albums spanning multiple genres, often prioritizing technical consistency over creative nuance^[60,61]. For example, classical or jazz tracks, which rely on subtle dynamic variations, may suffer from AI's aggressive compression^[60].

In mixing, AI serves as both a foundational tool and a collaborative partner. Tools such as Wave-U-Net and iZotope Neutron automate repetitive tasks like spectral masking, noise reduction, and vocal tuning^[62,63]. For amateurs, AI plugins offer “one-click” solutions to technical challenges, such as balancing instrument levels or reducing frequency clashes, democratizing access to professional workflows^[62]. Modified Wave-U-Net architectures, for instance, have shown promise in automatically

Table 1 Comparison of models from different periods.

Period	Type	Method	Remark
Before 2000	Statistical model	Markov chains	While effective for short-term dependencies, they lacked the capacity to capture long-term musical structures.
		Chomsky's grammars	Applicable to grammar-based and rule-based approaches, utilized predefined rules to generate melodies, offering stylistic consistency but limited adaptability.
		Genetic algorithms	It is less efficient but performs well in terms of exploration and diversity.
After 2000	Deep learning	RNN	RNN, particularly LSTM, enabled the modeling of temporal dependencies in music.
		CNN	For example, DeepBach incorporates past and future musical contexts, improving the coherence and musicality of generated melodies.
		GAN	For example, MuseGAN achieve breakthroughs in multi-track arrangement, BinaryMuseGAN improved note continuity.
		LLM	Integrating Large Language Models (LLM) into symbolic music generation to enhance melody, harmony, and arrangement processes.

mixing drum tracks with increased accuracy, rivaling human performance in dry mixing scenarios^[63]. Pro-am and professional users, however, prioritize AI as a starting point rather than a replacement. They use AI-generated suggestions for gain staging or EQ adjustments but refine results manually to align with artistic visions. While AI excels at technical tasks like error correction, it lacks the emotional and contextual understanding required for nuanced decisions, such as preserving the textural complexity of a jazz improvisation or the dynamic range of a symphonic piece^[58]. Professionals often integrate AI for efficiency in routine tasks (e.g., batch processing or stem separation) while retaining human control over creative elements like stereo imaging and emotional tone^[59,62].

Different user groups engage with AI tools in distinct ways. Amateurs rely on AI for accessibility, using platforms like LANDR to bypass technical barriers and achieve “radio-ready” masters with minimal expertise. For these users, AI serves as a substitute for formal training, enabling quick deployment on platforms like Spotify or Bandcamp^[57,62]. Pro-ams treat AI as a pedagogical aid, using tools to identify mixing flaws (e.g., phase issues or imbalanced bass) while retaining creative agency. They value AI’s ability to generate baseline mixes, which they then refine to match their artistic goals^[63]. Professionals adopt a hybrid approach, using AI for time-consuming technical tasks (e.g., loudness normalization or spectral analysis) while emphasizing human intuition for critical decisions^[59].

Recent advances have shown that LLM can orchestrate complex workflows in music production, audio processing, and mixing. One framework generates compositional audio using structured scripts, translating language prompts into sequences of model calls for generating speech, music, and effects, thereby enabling human-interpretable and modular audio creation^[64]. Another system expands on this by editing audio based on user instructions and existing clips through in-context learning and dialogue-based interactions^[65]. These LLM-powered agents support nuanced control, modular extensibility, and co-creative workflows, significantly enhancing the automation and adaptability of real-world audio production processes.

2.2.2 Audio synthesis and sound design

Deep learning architectures have enabled sophisticated audio generation. StyleGAN excels in synthesizing percussive and environmental sounds by learning latent space representations^[66,67]. Autoregressive models like WaveNet^[68] and AudioLM^[69] generate long-form musical sequences with coherent melodic and harmonic structures, while diffusion models (e.g., DifSound^[70]) produce high-fidelity environmental textures, surpassing GAN in timbral nuance. Multimodal models bridge semantic and acoustic domains. CLIP^[71] and its audio extension AudioCLIP^[72] enable text-conditioned sound synthesis. For example, using AudioCLIP to generate abstract sounds, finding that while cross-modal alignment remains limited, optimization of sinusoidal parameters yielded artistically engaging results, though constrained by the model’s inability to capture fine temporal dynamics^[73]. CLAP enhances audio-language mapping via contrastive learning, supporting zero-shot retrieval of sounds like “distant thunder” or “vibrant synth pads”^[74].

Interactive frameworks play a critical role in AI-assisted audio creation, enabling designers to exert more precise control over system outputs. Two representative systems have been proposed: one provides explicit control over domain-specific audio parameters, such as spectral centroid and attack-decay envelopes, while the other facilitates higher-level sonic manipulation through

latent space navigation based on the SeFa method^[67,75]. These systems are widely applied in real-time sound design contexts, such as adjusting timbral qualities and texture characteristics in game audio or film scoring, thereby enhancing the expressivity and adaptability of AI-generated sound. In the domain of music synthesis, a query-based concatenative synthesis framework has been developed, utilizing the Analytic Hierarchy Process (AHP) to assign differentiated weights to audio features (e.g., prioritizing pitch over loudness). This approach enables more intention-aligned material selection and reduces manual bias in the compositional workflow^[76]. In parallel, generative models are increasingly replacing traditional Foley recording methods, demonstrating high efficiency and realism in the automated synthesis of complex sound effects such as footsteps and fabric rustling^[77]. Furthermore, real-time audio processing tools—such as CataRT and Soundspotter—allow for dynamic concatenation and reorganization of audio segments, streamlining loop-based composition processes in genres like electronic music^[78,79].

A fundamental challenge lies in aligning AI-generated outputs with human auditory perception. Existing studies have identified limitations in AI timbre modeling, as outputs often exhibit excessive “metallic” artifacts or inconsistent temporal envelopes, indicating deficiencies in capturing complex auditory attributes such as harmonic overtones. To improve perceptual coherence in synthesized sequences, some approaches have introduced concatenation distance metrics to evaluate spectral continuity between segments, addressing the issue of homosonic segments—those acoustically similar yet perceptually distinct^[76]. User agency and ethical considerations are equally vital in system design. Research suggests that creators generally prefer AI as a collaborative assistant rather than a full substitute, and place high importance on precise control over key parameters, such as fade-in and fade-out curves^[73]. Ethical concerns include representational bias in training datasets—for example, the underrepresentation of non-Western instruments^[59]—as well as potential copyright risks associated with sample-based synthesis. To mitigate these issues, some systems incorporate transparency mechanisms, including explainable latent space editing^[80] and adjustable feature weighting^[78], thereby enhancing controllability, interpretability, and trustworthiness in creative applications.

In terms of LLM, a diffusion-based system enables text-to-sound generation, allowing users to synthesize and manipulate novel sounds directly within digital audio workstations, expanding beyond traditional libraries^[81]. Complementary work introduces audio-language models fine-tuned with dialogue-grounded data, enhancing co-creative mixing through contextualized feedback and instruction^[82]. These models facilitate intuitive interaction, dynamic sound generation, and pedagogical support. Collectively, they demonstrate the potential of AI to reshape sound design workflows by enabling both personalized creativity and efficient, collaborative production environments.

2.3 Music performance

2.3.1 Human-robot collaboration

AI-driven systems are reshaping musical collaboration through a spectrum of interaction models that encompass both passive and active engagement. One proposed hybrid framework conceptualizes the AI agent as a “supervisory-passive-active” entity capable of dynamically modulating parameters—such as percussive intensity and robotic gestures—to mirror human expressivity in real time^[83]. Central to such frameworks is the establishment of a shared mental model, as demonstrated by

systems that utilize Self-Organizing Maps (SOM) to learn patterns of human improvisation and generate context-aware musical responses^[84]. These developments are underpinned by the multimodal integration of sensor data and ML techniques. For example, gesture-based control systems employ Reinforcement Learning (RL) to adapt motion-to-sound mappings based on real-time user feedback—such as hand signals—to optimize responsiveness and musical relevance^[85]. Temporal coordination between human performers and AI agents has been further enhanced through oscillator-based synchronization models, enabling precise rhythmic alignment in live ensemble contexts^[86]. Such systems frequently adopt hybrid computational architectures that combine Deep Neural Networks (DNN) for pattern learning with rule-based modules for expressive control. A representative case involves LSTM trained on motion data to predict and modulate musical parameters dynamically.

A central theme in this domain is the perceived autonomy of AI agents. Empirical studies indicate that participants exhibit higher trust in AI systems that demonstrate proactive behavior, such as anticipating musical transitions during improvisation—an attribute exemplified in adaptive jazz performance systems^[87, 88]. Visual interpretability also plays a critical role: systems that visualize decision processes (e.g., mappings between movement and audio filters) enhance user comprehension and satisfaction, consistent with the “SimCity Effect”—the idea that transparency fosters trust. The fluidity of roles between the “tool” and “partner” paradigms is of critical importance in interactive music systems^[89]. Certain platforms feature AI agents trained on stylistic corpora—such as posthumous composers’ works—to act as co-performers, responding dynamically to live input while preserving stylistic fidelity^[90]. Grammar-based generative systems further complicate the boundary between algorithmic output and human creativity by emulating improvisational behavior in real time^[91].

Despite these advances, challenges related to authorship and bias remain salient. High-profile events, such as auctions involving AI-generated artworks, have underscored the legal ambiguities surrounding the use of copyrighted material in model training^[92]. Cultural biases embedded in training data—such as the underrepresentation of non-Western musical genres—may reinforce stereotypes, as reflected in critiques of improvisation systems that privilege male-centric archives^[89]. Striking a balance between machine agency and human intuition is equally imperative. Human-centered design frameworks argue that AI should augment, rather than replace, human creativity, and decision-making^[93]. Systems such as algorithm-assisted composition platforms maintain this balance by offering intelligent suggestions while preserving user control over creative direction^[86]. Moreover, latency in real-time performance—such as gesture-to-audio lag—necessitates the adoption of lightweight neural architectures on edge devices to ensure responsiveness.

The field is moving toward hybrid intelligent systems that deepen human-AI co-creation. Emerging paradigms, including quantum neural networks^[94] and collective robotic agents^[87], may enable distributed adaptive integration. Immersive technologies such as Virtual and Augmented Reality (VR/AR) can further facilitate real-time interaction with AI-generated virtual performers capable of responding simultaneously to both physical movement and affective cues^[95]. Long-term ethical frameworks must prioritize both transparency—through explainable AI mechanisms—and inclusive representation. Future co-creative systems should emphasize diverse training datasets and user-centered interaction design to ensure that AI enhances, rather than homogenizes, artistic expression^[96].

At present, LLM has realized human-machine music collaboration. One system leverages multi-modal inputs—including speech, gaze, and scene context—to guide physical robot performance through natural language instructions, synchronizing verbal and gestural outputs for expressive musical interaction^[97]. Another approach introduces dialog-based multi-robot coordination, where LLM agents plan, refine, and execute collaborative tasks through natural language dialogue and 3D spatial reasoning, adapting flexibly to real-world performance scenarios^[98]. Together, these frameworks demonstrate the potential of LLM to enhance co-creative agency, embodiment, and improvisation in interactive music performance systems.

2.3.2 Multimodal robot music performance

A core challenge in robotic musical performance lies in translating abstract emotional content into coherent bodily motion. One proposed cross-modal framework employs four parameters—Speed, Intensity, Regularity, and Extent (SIRE)—to synchronize emotional gestures, speech, and music^[99]. Experimental validation across all three modalities demonstrated that this four-dimensional representation could communicate basic emotions such as happiness and sadness with over 50% recognition accuracy, laying the groundwork for a unified model of affective expression. Building on this, a recent architecture integrates Gated Recurrent Units (GRU) for processing audio dynamics (e.g., rhythm and melody) with GoogleNet for extracting visual features such as facial expressions and stage lighting^[100].

Recent developments have further extended these frameworks through generative AI and LLM. For instance, the use of LLM such as GPT-3.5 Turbo has enabled the generation of robot choreography from natural language prompts, employing few-shot, zero-shot, and feedback-driven paradigms^[101]. This approach significantly enhanced audience-perceived artistic impact, highlighting the potential of language interfaces in improving robotic expressivity. Similarly, generative systems employing Cerebellar Model Articulation Controllers (CMAC) and multimodal reinforcement learning dynamically adjusted robotic trajectories based on musical structures (e.g., verse-chorus transitions) and stage cues, demonstrating high accuracy in motion planning across genres like classical and rock^[102].

Transformer-GAN architectures have also proven effective in generating visually synchronized performances: the GAN module produces realistic motion sequences while the Transformer captures temporal dependencies across audiovisual streams, resulting in notable improvements in rhythm-aligned choreography over baseline systems^[101, 103]. Multimodal Human-Robot Interaction (HRI) technologies are further enabling adaptive and emotionally resonant performances. Reinforcement learning agents trained with audience feedback have been shown to optimize gesture strength and spatial positioning, leading to a substantial increase in engagement compared to pre-scripted performances^[104].

The integration of computer vision and Natural Language Processing (NLP) has significantly enhanced responsiveness. For example, real-time facial recognition and voice emotion analysis allow robots to detect audience affect (e.g., excitement via smile detection) and modify musical output accordingly—such as switching from melancholic to cheerful melodies—while maintaining high accuracy in emotion mapping^[105]. In terms of expressive control, Gaussian process-based models have enabled nuanced piano dynamics based on tactile and visual sensing, emulating the interpretive qualities of classical human performers.

Moreover, for ensemble contexts, 3D-CNN-LSTM models have been employed to decode dynamic conducting gestures, allowing robotic performers to improvise in real time within jazz ensembles. This system demonstrated strong performance in gesture recognition, facilitating collaborative interaction with human musicians^[106].

2.4 Music analysis

2.4.1 Sentiment analysis

CNN and RNN play a pivotal role in the domain of audio-based emotion analysis. A CNN-LSTM hybrid architecture has been proposed to extract spectral-temporal features directly from raw audio signals, achieving effective classification of four emotional states—joy, sadness, anger, and calm—on the MAESTRO dataset^[107]. Another approach leverages 3D-CNN to model frequency-time-energy cubes, enabling the capture of hierarchical affective cues in orchestral music and yielding strong F1 performance on the EMODB dataset^[108]. In recent years, Transformer-based models have gained prominence. MusicBERT, a pre-trained model fine-tuned on millions of audio clips, employs self-attention mechanisms to model long-range dependencies in polyphonic music, achieving superior performance over LSTM baselines in cross-genre emotion recognition tasks^[109]. Graph Neural Networks (GNN) have also been introduced, wherein chord progressions are represented as topological graphs, and emotion inference is conducted via message-passing operations, yielding promising results on the MIREX dataset^[110].

A salient trend in current research is the integration of audio signals with multimodal musical metadata, such as lyrics, album information, and visual stimuli. Hierarchical multimodal fusion networks have been developed that combine BERT-based embeddings for lyric semantics with CNN-based representations of album artwork, resulting in high accuracy in emotion classification for popular music^[111]. Additionally, contrastive learning methods have been employed to align audio embeddings with Electroencephalogram (EEG) data, showing that shared latent representations significantly outperform unimodal models in emotion prediction. Transfer learning has proven effective in low-resource contexts; for instance, speech emotion recognition models pretrained on the IEMOCAP dataset have been successfully adapted to music via fine-tuning on annotated audio tracks, yielding performance comparable to models trained from scratch^[112]. Furthermore, zero-shot classification using pretrained audio transformers such as wav2vec 2.0 has demonstrated robust results in emotion recognition for underrepresented genres like traditional gamelan music^[113].

There is a growing emphasis on the psychological validity of computational models. Arousal-valence predictions validated against the Geneva Emotion Wheel exhibit strong alignment with human annotations, underscoring model reliability^[114]. To address cultural specificity, new models that incorporate melodic modes from diverse ethnic traditions have been shown to reduce bias in emotion recognition for non-Western music^[115]. Ethical concerns regarding representational bias in training data have also emerged. Analyses reveal that some models systematically overpredict “romantic” emotions for female artists, prompting the proposal of fairness-aware learning techniques such as reweighted loss functions to mitigate such disparities^[116]. In terms of practical applications, emotion recognition techniques have been deployed in music recommendation systems, music therapy, and interactive art. Emotion-aware recommender systems that incorporate collaborative filtering and emotion embeddings have significantly

enhanced user engagement in A/B testing environments^[117]. For live performance scenarios, an AI-driven orchestral conductor has been designed to dynamically adjust musical dynamics based on real-time facial emotion cues from the audience, leveraging a CNN-LSTM hybrid model for effective synchronization of emotional signals^[118].

LLM have significantly advanced music sentiment analysis by enabling nuanced and scalable emotional interpretation of lyrics. A hybrid approach integrating ChatGPT with a comprehensive Chinese emotion lexicon facilitated fine-grained sentiment detection across over 260 000 Mandarin pop lyrics (1967–2023), achieving 87% accuracy in binary sentiment prediction and surpassing lexicon-only methods by 23 percentage points. This method allowed multi-level emotion mapping—global sentiment, polarity intensities, primary emotions, and sub-emotions—revealing cyclical trends and archetypal emotional themes. Network analysis further highlighted how emotions co-occur and bridge sentiment polarities, underscoring LLM’ utility in affective cultural analytics^[119].

2.4.2 Form and harmony analysis

RNN and transformer architectures have become dominant in formal music analysis by modeling temporal dependencies in musical sequences. An LSTM-based model detects sectional boundaries (e.g., verse, chorus, and bridge) in pop music by analyzing spectro-temporal features, demonstrating high accuracy on the GTZAN dataset^[120]. A transformer-based model further enhances this capability by leveraging self-attention mechanisms to capture long-range structural patterns in orchestral music, improving segment classification performance relative to LSTM baselines^[121]. Other studies employ 3D-CNN to represent MIDI data as spatiotemporal grids, enabling hierarchical detection of phrase structures in piano works, achieving high F1-scores in sonata form analyses^[122, 123]. For open-form improvisation, transformer-based models have also been used to improve recognition of complex structural patterns^[124]. In addition, GNN have been applied to represent musical events as nodes in dynamic graphs, enabling the prediction of structural transitions—such as shifts from free jazz to swing—with strong classification performance^[123].

Harmonic analysis similarly benefits from ML’s capacity to model probabilistic relationships between chords and tonal contexts. A Bayesian network approach trained on a large corpus of annotated lead sheets can predict chord progressions in jazz standards and accurately generate typical harmonic patterns^[124]. A CNN-LSTM hybrid model processes audio spectrograms for real-time chord recognition, yielding high accuracy on the MAESTRO dataset^[125]. Contrastive learning techniques have been used to embed chords in a tonal space where Euclidean distance correlates with perceived harmonic tension, supporting semantic retrieval of harmonically similar passages—such as modal interchange pieces stylistically akin to the Beatles’ Yesterday^[126]. To address tonal ambiguity in post-tonal music, Gaussian Mixture Models (GMM) have been applied to model multi-key probability distributions, reducing classification error rates compared to single-label classifiers.

In recent years, multimodal analysis that integrates symbolic representations (e.g., MIDI and scores) with acoustic data (e.g., audio) has emerged as a prominent trend. A multimodal transformer model combines score annotations (such as tempo and articulation) with audio features to predict formal structures in classical symphonies, achieving high prediction accuracy^[127]. Cross-modal attention mechanisms have also been employed to

align harmonic changes with lyrical content in pop music, revealing a statistically significant correlation between minor key chord progressions and negative emotional sentiment in lyrics^[128]. Hierarchical attention networks have been proposed to model structural-harmonic interactions, with lower layers analyzing chord transitions and higher layers inferring formal sections. These models achieve strong performance in predicting harmonic cadence types—such as authentic and plagal cadences—at phrase boundaries in Baroque compositions^[129]. In terms of automated score parsing, OCR-CNN systems have demonstrated the ability to read handwritten scores, extract harmonic symbols, and classify formal structures, achieving high recall in Roman numeral harmonic analysis across multiple datasets^[130].

Recent advances have demonstrated that LLM exhibit partial competence in analyzing Western music harmony and form. While models such as GPT-3.5 and GPT-4.0 perform reliably on basic interval identification, they struggle with more complex chord and scale reasoning, particularly when enharmonic substitutions or inversions are introduced^[131]. Symbolic approaches using description logics and OWL ontologies further enable functional harmonic labeling and rule-based reasoning over musical progressions^[132]. These findings suggest a complementary path forward, integrating generative LLM with structured symbolic systems to enhance the interpretability and depth of automated music theory analysis.

3 Application of Artificial Intelligence in Music Recommendation and Consumption

3.1 Music recommendation

Early research on music recommendation systems primarily relied on Collaborative Filtering (CF) and content-based filtering. Singular Value Decomposition (SVD) was introduced to reduce the dimensionality of user-item interaction matrices for predicting individual music preferences^[133]. Subsequently, probabilistic Latent Semantic Analysis (pLSA) was applied to capture latent themes in user listening histories, thereby improving the quality of cross-gener recommendations and effectively reducing prediction error across multiple benchmark datasets^[134]. Content-based approaches focused on the extraction and modeling of audio features and metadata. For instance, GMM were employed to cluster music tracks based on timbral texture, rhythm, and tonality, enabling automatic recommendation of similar artists^[135, 136]. Additionally, Deep Belief Networks (DBN) outperformed traditional techniques such as Mel-Frequency Cepstral Coefficients (MFCC) in audio feature extraction, enhancing recommendation performance on large-scale music datasets^[136].

With the advancement of deep learning, CNN and RNN have become foundational architectures for sequential recommendation. Models such as GRU4Rec, built on Gated Recurrent Units (GRU), effectively process session-based listening sequences and outperform traditional CF-based methods in real-world applications^[137]. The Neural Factorization Machine (NFM) integrates factorization mechanisms with CNN to capture high-order user-item interactions, improving recommendation precision^[138]. The advent of the Transformer architecture, based on self-attention mechanisms, revolutionized long-range dependency modeling. The MusicTransformer model captures contextual relationships between songs in playlists and demonstrates superior ranking performance on mainstream music platforms^[139]. In addition, Vision Transformer (ViT) has been adapted to music recommendation by treating spectrograms as image-like inputs,

enabling cross-modal fusion of audio features and album cover visuals for enhanced multimodal recommendations^[140].

Contemporary systems increasingly incorporate multimodal data—including audio, lyrics, user reviews, and social network information—to enhance personalization and contextual awareness. For example, a hybrid recommendation model combining BERT-based lyric sentiment analysis with CNN-derived emotional audio features has been shown to significantly boost user engagement on major streaming platforms^[141]. Graph Convolutional Networks (GCN) have been applied to model user-artist-social connections in tripartite graphs, substantially improving recommendation diversity compared to unimodal baselines^[142, 143]. Cross-domain transfer learning techniques have been proposed to address data sparsity issues. Through domain adaptation, knowledge from domains such as movie rating prediction can be transferred to music recommendation tasks, yielding improved predictive performance in cold-start scenarios^[144, 145]. Furthermore, methods based on Pretrained Language Models (PLM) align user-generated reviews with audio features, enabling zero-shot recommendations for rare genres^[143].

Reinforcement Learning (RL) introduces a new paradigm for sequential decision-making in dynamic recommendation environments. Deep Q-Networks (DQN) have been utilized to optimize playlist continuation, dynamically adjusting recommendations based on real-time user feedback, leading to significantly longer session durations on commercial platforms^[146]. Policy gradient methods have also been applied to balance exploration and exploitation in music discovery, effectively reducing user churn^[147]. Increasing attention has been given to interactive recommendation interfaces. Conversational recommender systems built on LLaMA architectures allow users to express nuanced preferences through natural language, outperforming traditional keyword-based methods in terms of user satisfaction^[147, 148]. Moreover, the integration of haptic feedback into Virtual Reality (VR) recommendation environments enables users to “feel” rhythmic patterns via vibrotactile gloves, resulting in enhanced emotional engagement in preliminary user studies^[148].

Advanced recommendation models emphasize fine-grained user modeling and context awareness. Neural Collaborative Filtering (NCF) captures nonlinear user-item interactions through Multi-Layer Perceptrons (MLP), outperforming conventional matrix factorization approaches in recall-based metrics^[149]. The DeepFM model combines Factorization Machines (FM) for low-order feature interaction with CNN for high-order feature learning, yielding notable improvements in click-through rates for personalized playlists^[150]. Context-aware recommender systems incorporate temporal and situational signals. Models leveraging Temporal Convolutional Networks (TCN) identify daily listening patterns, optimizing music recommendations tailored to specific time windows^[151]. Additionally, frameworks that combine GNN with geolocation data enable environment-sensitive recommendations—such as suggesting reggae music for beach settings—achieving high predictive accuracy in spatially contextualized music consumption^[152].

3.2 Music consumption

AI-driven recommendation systems have become the core mechanism of contemporary music consumption, leveraging Collaborative Filtering (CF) and deep learning techniques to interpret user preferences. Research indicates that streaming platforms widely adopt CF-based algorithms to analyze user-item interaction matrices for optimizing content recommendations; however, such strategies tend to prioritize music released by major

record labels, potentially marginalizing independent artists' works^[153]. Matrix Factorization (MF) plays a pivotal role by projecting users and music into a shared latent space to predict personalized preference metrics such as play counts or listening duration. To address the dynamic nature of users' listening behaviors, scholars have proposed a hybrid recommendation framework combining collaborative filtering with Hidden Markov Models (HMM)—termed CF-HMM—which jointly models temporal evolution of user states and the mapping to song features, effectively enhancing recommendation accuracy on datasets like MSD^[154]. Moreover, this model incorporates context-aware mechanisms by integrating metadata (e.g., genre and artist) to further refine prediction precision.

The advent of deep neural networks has profoundly transformed the extraction of audio and contextual features. Among these, an Attention-Integrated Recurrent Neural Network (AIRNN), utilizing scattering transforms, extracts multi-scale audio features such as rhythm and timbre, substantially improving recommendation fidelity^[154]. By combining residual connections with multi-layer attention mechanisms, AIRNN weights sequential user interactions and surpasses baseline models like Music-CRN in performance. AI is also extensively applied to model user demographic attributes, historical behaviors, and real-time feedback (e.g., skips and likes) to generate personalized playlists^[155]. For instance, Spotify's Discover Weekly relies on such models to continually recommend niche music genres, significantly enhancing user engagement. Furthermore, CNN are widely employed to process album cover images and lyric texts, enabling cross-modal fusion of visual, linguistic, and audio preferences, thereby boosting the comprehensiveness and accuracy of recommendations.

AI systems exhibit robust capabilities in modeling complex user behaviors to predict music consumption patterns. Some studies critique existing recommendation algorithms for exacerbating the "filter bubble" effect, wherein popularity bias is reinforced by prioritizing content from established artists, limiting exposure for emerging talents^[153]. Nonetheless, such algorithms can also mine implicit user feedback (e.g., repeat plays and listening duration) to effectively identify long-tail content, enhancing diversity and discoverability. Literature categorizes AI roles in music recommendation into automation (e.g., chatbots responding to user queries) and augmentation (e.g., AI-assisted playlist curation)^[156]. Deep learning based recommender engines demonstrate high accuracy in predicting user preferences, while hybrid models that combine algorithmic efficiency with human curatorial intuition improve user experience without compromising system performance. Personalized recommendation not only reduces cognitive load but also facilitates seamless transitions between curated content and automated suggestions.

Ensuring trustworthiness and fairness in recommendation systems is essential for cultivating user confidence. Prior research has proposed trustworthy recommendation frameworks that integrate privacy-preserving techniques such as differential privacy alongside XAI. Utilizing Grad-CAM visualizations to highlight spectro-temporal audio features driving recommendations enhances system transparency and user satisfaction, demonstrating significant user experience benefits in musical electronic product applications^[154]. Additionally, AI can mitigate information asymmetry by employing Natural Language Processing (NLP) for sentiment analysis of user reviews, thereby refining lyric-based recommendations. Integration of real-time location data (e.g., GPS) facilitates context-aware

recommendations, such as delivering genre-appropriate music in specific environments, achieving excellent performance on real-world datasets^[155].

AI's impact further extends to operational efficiency and market dynamics. Studies report that algorithmically generated playlists constitute a major source of platform content, significantly increasing user engagement while reducing manual editorial effort^[153]. Nonetheless, copyright control by major record labels continues to influence recommendation system neutrality, underscoring the urgency for diversified data representation mechanisms. Further findings highlight that AI functionalities like predictive analytics and real-time feedback loops enable platforms to rapidly adapt to shifts in consumer preferences. For example, models detect seasonal trends such as heightened demand for festive music during holidays and dynamically adjust recommendations, resulting in increased playlist interactions during peak periods^[156].

4 Impact of Artificial Intelligence on the Music Industry

4.1 Structural transformations driven by AI

AI has profoundly transformed music creation by lowering technical barriers and fostering new modes of collaboration. Generative AI models such as DeepBach^[157] and MusicVAE^[158] have demonstrated the capability to produce compositions in distinct styles, spanning from Baroque chorales to contemporary pop melodies. These tools enable musicians to automate compositional tasks including harmonization and rhythm generation, serving as "creative assistants" rather than replacements^[158]. For instance, AI platforms like Amper Music allow users to generate customized soundtracks by selecting specific genres and emotional parameters, thereby significantly democratizing access to professional-quality production^[159]. However, this democratization prompts critical discussions regarding artistic authorship. Although AI generates raw material, human curation remains essential for imparting emotional depth and cultural context. As exemplified by certain collaborations, artists frequently assume the role of "editors" refining AI outputs and blurring the boundaries between human creativity and machine-generated content^[159]. This shift redefines the artist's role from sole creator to curator and collaborator, challenging conventional notions of artistic integrity.

Within production workflows, AI optimizes efficiency through automated mixing, mastering, and sound design. Tools like LANDR employ machine learning to analyze audio quality and propose enhancements, thereby reducing reliance on specialized audio engineers^[160]. For independent musicians and small labels, such AI technologies level the competitive landscape, enabling high-quality production at reduced costs. Nonetheless, the tendency of AI to replicate dominant styles present in its training data poses risks of stylistic homogenization and potential inhibition of innovation^[161].

AI-driven recommendation systems deployed by platforms such as Spotify and YouTube have revolutionized music discovery. Utilizing collaborative filtering and content-based analysis, these systems personalize playlists and enhance user engagement, yet they also generate "filter bubbles" that prioritize popular genres while marginalizing niche and experimental music^[162]. Algorithmic biases may inadvertently reinforce cultural stereotypes by favoring Western popular music over non-Western traditions, thereby threatening global musical diversity^[163]. Furthermore, the proliferation of AI-generated content

complicates distribution ecosystems. Large quantities of AI-composed music are hosted on major platforms, often without explicit labeling, complicating differentiation from human-created works and raising concerns regarding transparency and user awareness^[159]. Concurrently, decentralized models such as NFTs and blockchain-based platforms (e.g., Audius) are emerging as alternatives that empower independent artists to bypass traditional intermediaries and directly monetize their content, albeit facing challenges related to scalability and regulatory clarity^[164].

AI is reshaping the economic landscape of the music industry by altering revenue distribution models. Streaming platforms increasingly rely on AI to optimize royalty allocation and targeted advertising, yet these systems disproportionately benefit major artists and labels. Independent creators often face difficulties in securing equitable compensation, as algorithms emphasize high-streaming content, resulting in a “long tail” effect where only a small fraction of creators thrive^[163, 165]. Concurrently, new revenue streams have emerged via AI-generated commercial music used in film, gaming, and advertising. Companies such as AIVA and Amper Music provide licensable AI compositions to meet industry demand for affordable and customizable content^[158]. However, these developments raise complex questions regarding copyright ownership; current legal frameworks, which require human authorship for copyright protection, struggle to accommodate AI-generated works, creating a legal gray area^[166]. Moreover, AI is transforming Artist and Repertoire (A&R) processes. Certain platforms leverage machine learning to identify emerging talent by analyzing streaming metrics and social media engagement, thereby reducing dependence on traditional human scouts^[160]. While this democratizes A&R, it also risks privileging data-driven trends over artistic innovation, potentially fostering formulaic music production.

Generative AI is reshaping employment, income, and task composition across music occupations through both substitution and augmentation. Macro-evidence indicates no aggregate fall in labour demand so far; rather, AI is altering job quality, reallocating tasks, and improving matching and productivity in creative labour markets, implying near-term employment effects that are modest but heterogeneous across roles^[167]. In music specifically, a large cross-national study commissioned by CISAC projects substantial income pressure: by 2028, 24% of music creators’ revenues are “at risk” as AI outputs compete with human works and as unlicensed training transfers value to model providers; streaming and library markets are predicted to see the highest substitution^[168]. Micro-evidence from a 2023–2024 survey of 15 000 German and French Collecting-Society Members (GEMA/SACEM) indicates functional reconfiguration: creators view AI most negatively for core creative tasks (56% negative) but more positively for supporting tasks (e.g., marketing and promotion), with higher uptake among younger cohorts—patterns consistent with a bifurcation between human-led creation and AI-enabled production support^[169]. Sectoral statistics underscore the employment stakes: the UK music industry employed approximately 216 000 people in 2023 while characterizing generative AI as a “wild west” challenge demanding policy response, suggesting distributional impacts on session musicians, library composers, and post-production technicians may precede aggregate job losses^[170].

4.2 Ecological challenge in the AI-driven music industry

The advanced integration of AI into music creation has revealed fundamental gaps within existing copyright law. Traditional legal frameworks primarily assign authorship and ownership rights to

natural persons, yet the legal status of AI-generated works remains ambiguous. Under current regulations, training AI systems on copyrighted audio data without authorization may constitute infringement and trigger litigation, particularly when generative models extensively utilize protected musical works to “learn” styles and structures^[160]. Courts have not yet recognized AI as a legal subject capable of authorship, which results in a legal vacuum where ownership of AI-generated content remains undefined. This lack of legal clarity hampers the commercialization of such content and undermines investor confidence in AI-driven creative industries^[163].

In addition to legal uncertainty, the use of AI in cross-cultural musical contexts raises pressing ethical concerns, especially regarding cultural appropriation. AI models are frequently trained on vast, heterogeneous datasets that include a significant number of non-Western musical elements. However, in the absence of transparent attribution or statements of cultural provenance, AI-generated music may exploit these resources without proper contextualization. For instance, a model trained on African drumming traditions may produce commercially viable music that mimics these patterns without acknowledging their cultural origins, potentially leading to misrepresentation or even distortion of cultural meanings^[161]. In light of increasing digitalization and automation in the global music industry, the development of ethical guidelines has become essential to safeguard cultural intellectual property and intangible heritage.

The influence of AI on musical aesthetics is also significant. Given that generative systems rely heavily on historical datasets, they tend to reproduce dominant stylistic norms, thereby reinforcing the prevalence of Western pop and electronic genres^[162]. Such algorithmic bias risks marginalizing niche, regional, or traditional musical forms in the digital landscape. In terms of musical attributes such as modality and rhythmic structure, AI systems are more adept at replicating frequently occurring patterns in training data, while struggling to handle “non-standard” musical features like microtonality or irregular rhythmic schemes^[161]. Over time, this may erode musical diversity and reduce the cultural inclusivity of music creation.

Despite technical sophistication, AI still lacks the capacity to authentically convey the emotional and contextual depth that characterizes human musical expression. Music is not merely a sequence of organized sounds but a medium through which emotional experiences, personal narratives, and cultural identities are articulated. Lacking lived experience, affective sensitivity, and cultural intuition, AI-generated compositions often fail to embody the nuanced meaning and emotional resonance that human creators imbue in their work^[171]. As a result, the authenticity of AI-generated music continues to be contested, particularly regarding its ability to reflect unique artistic perspectives and represent cultural identities.

AI’s entry into the music production process has transformed the distribution of roles within the technical workforce. Tools that automate mixing, mastering, and sound design have reduced reliance on audio engineers and accelerated production efficiency. While some roles are at risk of being displaced, professionals who embrace AI as a supportive tool can achieve greater creative freedom and productivity. For example, many producers now use AI to manage repetitive tasks such as audio cleanup or tonal balancing, freeing up time for higher-order creative decision-making, including stylistic direction and emotional articulation^[160].

Moreover, the rise of AI in music curation and data analytics has generated demand for hybrid professional roles that blend artistic judgment with technological literacy. However,

independent artists are increasingly confronted with dual pressures: the proliferation of low-cost, high-output AI-generated content flooding distribution platforms, and the challenge of maintaining artistic distinctiveness in an algorithmically saturated ecosystem. To remain competitive, these artists must develop personalized stylistic signatures and emphasize cultural specificity. Notably, AI also opens new avenues for collaboration. In hybrid creative models, artists may use AI to generate initial ideas, which are then refined through manual aesthetic intervention and emotional reinterpretation^[158]. Ultimately, the key challenge lies in achieving a dynamic balance between the efficiencies of automation and the preservation of human artistry, in order to sustain the creative vitality and cultural integrity of the music industry.

4.3 Opportunity for innovation and balance

The future of AI in the music field increasingly points toward collaborative models that integrate the efficiency of machines with the originality and emotional depth of human creativity. Rather than positioning AI as a replacement for musicians, emerging frameworks emphasize augmentation—tools that enhance artistic potential rather than diminish it. A notable example is the Flow Machines project, which allows composers to guide AI algorithms in generating musical ideas, facilitating a form of “co-authorship” that preserves human agency while expanding the boundaries of musical expression^[172]. These systems enable artists to explore complex harmonic structures, unexpected stylistic fusions, and novel sonic textures that might not be readily conceived through conventional methods.

Technological convergence further amplifies these possibilities. The integration of AI with immersive media platforms such as Virtual Reality (VR) and Augmented Reality (AR) promises to reshape both music creation and performance. For instance, AI-generated soundscapes embedded in VR environments can create deeply interactive concert experiences where audience members are not passive observers but active participants navigating dynamic auditory and visual landscapes. Such innovations offer transformative potential for live events, enabling performances that transcend the spatial and sensory limitations of traditional venues^[162].

However, the realization of this collaborative vision requires significant legal, ethical, and infrastructural developments. The current lack of legal clarity surrounding the ownership of AI-generated music constitutes a major barrier to industry adoption and innovation. Traditional copyright regimes are ill-equipped to assign rights in works produced by non-human agents. To address this, policymakers must establish comprehensive frameworks that explicitly define the ownership, attribution, and liability associated with AI-generated works. One possible approach is the mandatory disclosure and labeling of AI-produced content, which would enhance transparency and empower consumers to make informed choices regarding authenticity and authorship^[160].

Moreover, the need for international harmonization of copyright law has become increasingly pressing. As AI-generated music crosses borders and digital platforms operate on a global scale, inconsistencies in legal treatment across jurisdictions can hinder fair compensation and rights protection. A coordinated global effort is essential to ensure that human contributors are properly credited and remunerated, while also safeguarding intangible cultural heritage from appropriation or misuse^[166]. In addition to legal reform, technical and cultural measures are

equally critical. Training datasets used in generative music systems must be diversified to include non-Western, indigenous, and marginalized musical traditions. Overreliance on Western-centric data can perpetuate stylistic homogenization and cultural erasure. Collaborative efforts among technologists, ethnomusicologists, and practitioners can ensure that AI tools are designed with cultural sensitivity and foster rather than diminish musical diversity^[161]. In this way, AI can evolve not as a homogenizing force, but as a medium for cross-cultural dialogue and innovation.

National approaches to AI in music are converging on two themes—human authorship and transparency—yet diverge on training, registration, and voice rights. The EU AI Act imposes transparency for synthetic content relevant to musical outputs^[173]. In the United States, the Copyright Office and the D.C. Circuit reaffirm that purely machine-generated works lack copyright, while AI-assisted works are registrable only where human creativity is evident; states such as Tennessee prohibit unauthorized voice cloning (ELVIS Act)^[174]. The United Kingdom has no bespoke statute; government consultation continues on transparency and Text-and-Data Mining (TDM)^[175]. Japan’s Copyright Act Article 30-4 broadly permits “information analysis” (TDM), but guidance cautions that style-targeted training may fall outside the exception^[176]. Republic of Korea’s collecting society KOMCA now refuses registration and royalties for songs involving AI, constraining commercialization despite no statutory ban^[177]. China’s deep-synthesis and generative-AI rules require labeling and platform duties; 2025 measures add national AIGC labeling^[178]. Singapore provides a counterpoint: the 2021 Copyright Act’s computational-data-analysis exception permits TDM on lawfully accessed materials, and its Model Governance Framework encourages provenance and watermarking^[179]. Overall, countries share a human-authorship premise and trend toward transparency, but differ on training permissibility, collective-management acceptance, and the legal status of voice cloning, and enforcement remains uneven globally.

Ultimately, the key to sustainable development in AI-driven music lies in striking a dynamic balance: leveraging automation without sacrificing human craftsmanship, embracing innovation while preserving cultural and artistic integrity. Only through this balanced approach can the industry fully realize AI’s creative potential.

5 Artificial Intelligence and Interdisciplinary Research

5.1 Music education

AI is transforming music education from traditional uniform instruction to precision-based learning through data-driven personalization. The ITMMCAI-MCA-ACNN system, for instance, incorporates an Attention-based Convolutional Neural Network (ACNN) optimized by the Musical Chairs algorithm to effectively isolate audio components—such as drums and vocals—from mixed signals^[180]. This enables students to concentrate on specific instrumental or rhythmic layers for targeted practice, thereby enhancing their rhythmic perception and pitch discrimination. Similarly, the EFDFO framework leverages whale optimization algorithms to analyze integrated student data, including performance indicators and biometric responses, in order to generate individualized pedagogical strategies^[181]. These systems offer real-time corrective feedback—such as the automatic detection and adjustment of intonation or rhythmic inaccuracies—addressing the inherent

limitations of conventional one-size-fits-all teaching models^[183].

VR technology further expands the potential of personalized instruction by creating immersive environments for experiential learning. A VR-based vocal training system simulates performance scenarios, allowing students to practice breath control and stage presence in a safe, controlled setting^[183]. Moreover, 3D visualizations of vocal mechanisms render abstract vocal techniques more accessible and comprehensible, thus overcoming the visualization barriers often encountered in traditional instruction. Additionally, AI-powered Information and Communication Technology (ICT) tools—such as interactive rhythm modules—adapt learning materials dynamically based on each student's individual progress, significantly improving engagement and effectiveness in remote music education^[184].

AI also plays a critical role in advancing accessible and inclusive music education. GCompris, a free multilingual platform, offers interactive activities such as melody recognition and rhythm-based games, accommodating diverse learning styles, including visual and kinesthetic modalities^[185]. This aligns with UNESCO's agenda for inclusive education by ensuring a broad range of learners can participate meaningfully. For students with disabilities, AI-driven transcription tools, such as automatic audio-to-notation converters, enable the transformation of auditory input into visual output, thereby promoting equal access to music learning environments^[186]. In regions with limited educational infrastructure, initiatives like the Enferm_Arte project employ AI to analyze historical artworks portraying diseases, integrating cross-disciplinary content into music curricula to contextualize cultural narratives and enhance local cultural awareness^[187].

AI further empowers students to explore music composition and improvisation. Tools such as Amper Music and AIVA generate harmonic progressions and melodic patterns that provide structurally coherent and theory-informed resources, inspiring creativity among novice composers^[181]. Experimental educational projects have also adopted AI to facilitate collaborative sound art creation, where students use AI-processed stone sculptures to produce generative works combining acoustic textures with digital processing techniques^[188]. These methods reflect the principles of the "slow movement" in education, emphasizing experiential and reflective learning over rote memorization^[189]. Collaborative AI platforms—including low-latency virtual ensembles—enable synchronous global musical collaboration. Such systems allow remote participants to share high-fidelity audio in real time, fostering intercultural exchange and creative synergy across borders^[190]. The Falerone Art Colony project exemplifies this integration, with architecture and music students employing AI to develop site-specific installations that blend technical competence with artistic expression^[191].

Despite these advancements, AI in music education still encounters technical and pedagogical limitations. While algorithms exhibit impressive precision in audio analysis, they remain limited in interpreting the emotional nuances intrinsic to musical expression^[192]. Moreover, excessive reliance on intelligent systems, if not balanced with traditional methods, may erode essential foundational skills such as manual notation and aural training, thereby affecting students' long-term musical development^[193]. Ethical concerns are also emerging, particularly regarding data privacy and the risk of cultural homogenization. While AI enhances the efficiency of evaluative processes—such as automated performance assessment—it may also marginalize culturally specific musical practices, posing a threat to the preservation of diverse musical heritages^[194]. As highlighted in recent studies on Lithuanian rural music, there is a pressing need

to develop culturally responsive AI systems that are sensitive to local traditions rather than enforcing standardized models^[195].

5.2 Music therapy

AI's ability to analyze vast datasets and generate personalized musical interventions is fundamentally transforming how individualized needs are addressed in music therapy. For example, the Melomics-Health algorithm composes music tailored to specific therapeutic goals such as stress reduction or pain relief by modulating parameters like timbre, tempo, and harmonic structure, thus producing standardized yet adjustable auditory stimuli^[196]. In a randomized controlled trial, participants who listened to AI-generated music exhibited relaxation levels comparable to those who listened to self-selected music, as measured by the Visual Analogue Scale (VAS). The integration of neural networks further facilitates personalized interventions by decoding physiological and neural signals to inform therapeutic strategies. A Deep Belief Network (DBN) has been employed to classify EEG signals from patients with depression during music therapy, effectively distinguishing between EEG patterns elicited by different musical styles (e.g., upbeat music vs. dirges), thereby assisting therapists in emotional state modulation^[197]. Similarly, an optimized LSTM was used to analyze music therapy data from bereaved families, achieving strong performance in predicting psychological trauma recovery and identifying individuals who responded positively to AI-personalized music interventions^[198].

AI also offers novel insights into the interaction between music and the brain. Machine learning techniques have been utilized to model correlations between physiological signals (e.g., electrodermal activity and blood volume pulse) and emotional responses elicited by various music genres^[199]. Neural networks have demonstrated high accuracy in genre classification and emotion prediction based on subjective assessments, revealing the neural correlates underlying music perception. Functional neuroimaging studies have shown that AI-generated music activates reward-related brain structures (e.g., the nucleus accumbens) and modulates stress-associated regions (e.g., the hypothalamus), paralleling the neural responses induced by human-composed music, but with greater consistency in parameter control^[200]. The classical "Mozart Effect" which posits that listening to classical music enhances cognitive performance, has been reinterpreted through an AI lens. AI-generated music incorporating specific frequency bands (e.g., gamma waves to promote focus) has been shown to replicate the cognitive benefits associated with Mozart's works while reducing cultural bias through data-driven genre selection. This challenges traditional assumptions that music therapy relies on familiar repertoires, suggesting AI's capacity to produce novel, universally effective auditory stimuli^[200].

Conventional music therapy often faces challenges regarding scalability and standardization. AI addresses these gaps by automating music generation and providing quantifiable metrics. Co-design workshops with therapists have highlighted key barriers to AI integration, including resistance to technology and the need to optimize hybrid human-AI collaboration models^[201]. The implementation of AI tools for real-time emotion recognition during therapy sessions enables dynamic adjustment of interventions by therapists. For populations with limited access to trained professionals—such as those in rural or aging communities—AI offers scalable alternatives. For instance, AI-driven robotic systems that simulate live music performances have been used to provide rhythmic and melodic cues aimed at improving motor coordination in patients with dementia^[202].

These systems, equipped with sensors to monitor engagement, have demonstrated greater patient involvement compared to passive listening sessions.

Despite improvements in therapeutic precision, the application of AI in music therapy raises ethical and technical concerns. There is a risk of algorithmic bias in music generation, necessitating the use of diverse training datasets to avoid reinforcing cultural or emotional stereotypes^[196]. Additionally, therapists have expressed concerns about the potential erosion of the relational core of therapy if AI were to replace human interaction. Accordingly, AI should be designed as a “collaborative tool” that supports, rather than substitutes, functions such as administrative tasks or pre-session planning^[201]. Technical limitations persist, including the high computational demands of real-time neural modeling. Training DBN requires large EEG datasets, which may be difficult to obtain in resource-constrained settings^[197]. Future research must therefore prioritize the development of lightweight models and edge computing strategies to facilitate AI deployment in low-resource environments.

6 Research Conclusion and Future Research Direction

6.1 Research conclusion

This study systematically reviewed the multifaceted applications of AI in the field of music, encompassing key domains such as music composition, audio production, performance collaboration, emotional analysis, form and harmony analysis, as well as music recommendation and consumption. The findings demonstrate that AI technologies—particularly deep learning, reinforcement learning, generative models, and multimodal fusion mechanisms—have profoundly reshaped the modes of music creation and dissemination. On one hand, AI has significantly enhanced the automation and personalization of music content generation, achieving notable improvements in both efficiency and quality across melody generation, lyric writing, mastering, and mixing optimization. On the other hand, AI-driven recommendation systems, based on collaborative filtering and context-awareness, enable fine-grained modeling of user preferences, effectively increasing user engagement and platform retention. Furthermore, AI exhibits strong auxiliary capabilities in interdisciplinary fields such as music education and music therapy, thereby advancing the social functions of music. Overall, AI not only augments the intelligence of music systems at the technical level but also triggers profound impacts across artistic, economic, and ethical dimensions, reflecting a pronounced structural transformation.

6.2 Future research direction

Despite significant breakthroughs, AI in music still faces several critical challenges and unresolved issues. Future research may be directed toward the following areas.

(1) Enhancing model explainability and emotional expressiveness. AI models in music generation and recommendation often suffer from the “black box” problem. In such cases, the internal decision-making processes are opaque to both users and developers. This lack of transparency undermines user trust and limits the interpretability necessary for clinical and creative applications. Future research should prioritize the development of XAI frameworks. These frameworks must offer transparent model architectures and intuitive interfaces. They should also enable interpretable representations of latent spaces,

timbral parameter adjustments, and recommendation logic. In particular, explainability mechanisms must align semantics, rhythm, and emotional cues within lyric generation and recommendation systems. Doing so ensures that outputs remain congruent with users’ psychological and cultural contexts. Moreover, current AI systems fall short in simulating authentic human emotional expression in music. Subsequent work could integrate multimodal emotion recognition—including facial expressions, vocal intonations, and neurophysiological feedback. This would elevate the expressiveness and emotional depth of generated music, fostering more human-like affective experiences.

(2) Improving adaptability to non-western and cross-cultural musical styles. Most existing AI music models are trained predominantly on Western popular music datasets. This results in stylistic homogenization and limited cultural representation. Future work must focus on building inclusive corpora that encompass minority, regional, and non-Western musical traditions. Incorporating ethnomusicological theories and advanced semantic embedding techniques can enhance cultural sensitivity. These methods will help develop analytical and generative models that are inclusive and expressive. They also mitigate the risks of “technological colonialism”, where dominant cultural paradigms marginalize minority traditions. Furthermore, cross-cultural adaptability research should explore how AI captures and conveys culturally specific emotional and symbolic meanings. This would support global musical exchange and promote intercultural dialogue.

(3) Developing collaborative mechanisms between AI and human creativity. As generative models and assistive tools evolve, the role of AI in music creation grows. However, defining the boundaries and modalities of human-AI collaboration remains a major challenge. Future research should focus on creating interactive co-creation platforms within a hybrid intelligence framework. These platforms should enable musicians to regulate AI involvement and stylistic influence. In doing so, they help preserve individual artistic identity and originality. Attention should also be given to real-time feedback and user experience design. Systems must respond to musicians’ creative intent and emotional state in real time. Intuitive interfaces and transparent decision pathways are essential for establishing AI as an intelligent creative partner—rather than a replacement.

(4) Institutionalizing legal and ethical frameworks. The rise of AI-generated music raises unprecedented legal and ethical concerns. These include ownership, authorship, and use transparency. Future studies must work with legal scholars and policymakers to develop robust frameworks tailored to AI music creation. Such frameworks should include systems for provenance tracking and attribution. They must also ensure the ethical use of training data and establish fair copyright and revenue-sharing mechanisms. Ethical considerations must be embedded throughout the technology lifecycle. This includes employment displacement, artistic freedom, creator identity, algorithmic bias, and cultural stereotyping. Responsible and sustainable deployment depends on these safeguards.

(5) Lightweight deployment via edge computing and expansion into educational contexts. Many current AI music systems depend on high-performance computing infrastructure. This restricts their use in resource-constrained settings. Future research should focus on lightweight neural network design and edge computing strategies. These will enable local, real-time use of AI music tools. Such advancements reduce reliance on cloud services, improve latency, protect privacy, and lower access barriers. This is especially important for underserved regions,

developing nations, and marginalized communities. In parallel, AI music systems can be integrated into education. Personalized learning tools and adaptive skill-building platforms can democratize music education. This supports cultural dissemination and promotes social inclusion.

Dates

Received: 25 June 2025; Revised: 18 September 2025; Accepted: 21 September 2025

References

- [1] Y. Chai, C. Miao, B. Sun, Y. Zheng, and Q. Li, Crowd science and engineering: Concept and research framework, *International Journal of Crowd Science*, vol. 1, no. 1, pp. 2–8, 2017.
- [2] S. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach (4th edition)*. London, UK: Pearson, 2021.
- [3] Y. LeCun, Y. Bengio, and G. Hinton, Deep learning, *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [4] D. Cope, Experiments in musical intelligence (EMI): Non-linear linguistic-based composition, *Interface*, vol. 18, nos. 1&2, pp. 117–139, 1989.
- [5] J. P. Briot, G. Hadjeres, and F. D. Pachet, *Deep Learning Techniques for Music Generation*. Cham, Switzerland: Springer, 2020.
- [6] P. Dhariwal, H. Jun, C. Payne, J. W. Kim, A. Radford, and I. Sutskever, Jukebox: A generative model for music, arXiv preprint arXiv: 2005.00341, 2020.
- [7] D. Herremans, C. H. Chuan, and E. Chew, A functional taxonomy of music generation systems, *ACM Comput. Surv.*, vol. 50, no. 5, pp. 1–30, 2018.
- [8] L. Smith, Agency & autonomy: Intersections of artificial intelligence and creative practice, <https://www.academia.edu/95498112>, 2020.
- [9] F. P. Brooks, A. L. Hopkins, P. G. Neumann, and W. V. Wright, An experiment in musical composition, *IRE Transactions on Electronic Computers*, vol. EC-6, no. 3, pp. 175–182, 1957.
- [10] S. R. Holtzman, A generative grammar definition language for music, *Interface*, vol. 9, no. 1, pp. 1–48, 1980.
- [11] S. R. Holtzman, Using generative grammars for music composition, *Comput. Music. J.*, vol. 5, no. 1, p. 51, 1981.
- [12] B. Lindblom and J. Sundberg, *Towards a Generative Theory of Melody*. Stockholm, Sweden: Royal Swedish Academy of Music, 1970.
- [13] J. Biles, GenJam: A genetic algorithm for generating jazz solos, in *Proc. 19th international computer music conference (ICMC)*, Tokyo, Japan, 1994, pp. 131–137.
- [14] D. Eck and J. Schmidhuber, A first look at music composition using LSTM recurrent neural networks, Tech. Rep. No. IDSIA-07-02, Istituto Dalle Molle Di Studi Sull Intelligenza Artificiale, Manno, Switzerland, 2002.
- [15] E. Waite, Generating long-term structure in songs and stories, <https://magenta.tensorflow.org/2016/07/15/lookback-rnn-attention-rnn>, 2016.
- [16] F. Colombo, S. Alexander, and G. Wulfram, Deep artificial composer: A creative neural network model for automated melody generation, presented at the Computational Intelligence in Music, Sound, Art and Design: 6th International Conference, Amsterdam, the Netherlands, 2017.
- [17] H. Chu, R. Urtasun, and S. Fidler, Song from PI: A musically plausible network for pop music generation, arXiv preprint arXiv: 1611.03477, 2016.
- [18] A. Roberts, J. Engel, C. Raffel, C. Hawthorne, and D. Eck, A hierarchical latent vector model for learning long-term structure in music, arXiv preprint arXiv: 1803.05428, 2018.
- [19] C. Z. A. Huang, A. Vaswani, J. Uszkoreit, I. Simon, C. Hawthorne, N. Shazeer, A. M. Dai, M. D. Hoffman, M. Dinculescu, and D. Eck, Music transformer, arXiv preprint arXiv: 1809.04281, 2018.
- [20] G. Medeot, S. Cherla, K. Kosta, M. McVicar, S. M. Abdallah, M. Selvi, E. Newton-Rex, and K. Webster, StructureNet: Inducing structure in generated melodies, in *Proc. International Society for Music Information Retrieval Conference*, Paris, France, 2018, pp. 725–731.
- [21] K. Chen, W. Zhang, S. Dubnov, G. Xia, and W. Li, The effect of explicit structure encoding of deep neural networks for symbolic music generation, in *Proc. Int. Workshop on Multilayer Music Representation and Processing (MMRP)*, Milan, Italy, 2019, pp. 77–84.
- [22] C. F. Huang, Y. S. Lian, W. P. Nien, and W. H. Chieng, Analyzing the perception of Chinese melodic imagery and its application to automated composition, *Multimedia Tools and Applications*, vol. 75, no. 13, pp. 7631–7654, 2016.
- [23] G. Hadjeres and F. Nielsen, Interactive music generation with positional constraints using anticipation-RNNs, arXiv preprint arXiv: 1709.06404, 2017.
- [24] B. L. Sturm, J. F. Santos, O. Ben-Tal, and I. Korshunova, Music transcription modelling and composition using deep learning, arXiv preprint arXiv: 1604.08723, 2016.
- [25] H. W. Dong, W. Y. Hsiao, L. C. Yang, and Y. H. Yang, MuseGAN: Multi-track sequential generative adversarial networks for symbolic music generation and accompaniment, presented at the 32nd AAAI Conference on Artificial Intelligence, New Orleans, LA, USA, 2018.
- [26] H. W. Dong and Y. H. Yang, Convolutional generative adversarial networks with binary neurons for polyphonic music generation, arXiv preprint arXiv: 1804.09399, 2018.
- [27] H. M. Liu and Y. H. Yang, Lead sheet generation and arrangement by conditional generative adversarial network, in *Proc. 17th IEEE Int. Conf. Machine Learning and Applications (ICMLA)*, Orlando, FL, USA, 2018, pp. 722–727.
- [28] M. Scirea, J. Togelius, P. Eklund, and S. Risi, MetaCompose: A compositional evolutionary music composer, in *Evolutionary and Biologically Inspired Music, Sound, Art and Design*, C. Johnson, V. Ciesielski, J. Correia, and P. Machado, eds. Cham, Switzerland: Springer International Publishing, 2016, pp. 202–217.
- [29] R. De Prisco, G. Zaccagnino, and R. Zaccagnino, EvoComposer: An evolutionary algorithm for 4-voice music compositions, *Evol. Comput.*, vol. 28, no. 3, pp. 489–530, 2020.
- [30] M. A. Kaliakatsos-Papakostas, A. Floros, and M. N. Vrahatis, Interactive music composition driven by feature evolution, *SpringerPlus*, vol. 5, no. 1, p. 826, 2016.
- [31] X. Liang, X. Du, J. Lin, P. Zou, Y. Wan, and B. Zhu, ByteComposer: A human-like melody composition method based on language model agent, arXiv preprint arXiv: 2402.17785, 2024.
- [32] L. Ou, J. Zhao, Z. Wang, G. Xia, Q. Liang, and T. H. Y. Wang, Unifying symbolic music arrangement: Track-aware reconstruction and structured tokenization, arXiv preprint arXiv: 2408.15176, 2024.
- [33] X. Zhang and M. Lapata, Chinese poetry generation with recurrent neural networks, in *Proc. 2014 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, Doha, Qatar, 2014, pp. 670–680.
- [34] M. Ghazvininejad, X. Shi, Y. Choi, and K. Knight, Generating topical poetry, in *Proc. 2016 Conf. Empirical Methods in Natural Language Processing*, Austin, TX, USA, 2016, pp. 1183–1191.
- [35] E. Manjavacas, M. Kestemont, and F. Karsdorp, Generation of hip-hop lyrics with hierarchical modeling and conditional templates, in *Proc. 12th Int. Conf. Natural Language Generation*, Tokyo, Japan, 2019, pp. 301–310.
- [36] E. Malmi, P. Takala, H. Toivonen, T. Raiko, and A. Gionis, DopeLearning: A computational approach to rap lyrics generation, in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 195–204.
- [37] Y. Tian, A. Narayan-Chen, S. Oraby, A. Cervone, G. Sigurdsson, C. Tao, W. Zhao, T. Chung, J. Huang, and N. Peng, Unsupervised

- melody-to-lyrics generation, in *Proc. 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Toronto, Canada, 2023, pp. 9235–9254.
- [38] Z. Sheng, K. Song, X. Tan, Y. Ren, W. Ye, S. Zhang, and T. Qin, SongMASS: Automatic song writing with pre-training and alignment constraint, *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 15, pp. 13798–13805, 2021.
- [39] M. Güven, Why “sway” again? Prosodic constraints and singability in song (re)translation, in *Studies from a Retranslation Culture*, Ö. B. Albachten and Ş. T. Gürçağlar, eds. Singapore: Springer, 2019, pp. 177–194.
- [40] M. Khoshsaligheh and S. Ameri, Exploring the singability of songs in *A Monster in Paris* dubbed into Persian, *Asia Pac. Transl. Intercult. Stud.*, vol. 3, no. 1, pp. 76–90, 2016.
- [41] Q. Liang, X. Ma, F. Doshi-Velez, B. Lim, and Y. Wang, XAI-lyricist: Improving the singability of AI-generated lyrics with prosody explanations, in *Proc. 33rd International Joint Conference on Artificial Intelligence*, Macao, China, 2023, pp. 7877–7885.
- [42] Z. Hong, C. Cui, R. Huang, L. Zhang, J. Liu, J. He, and Z. Zhao, UniSinger: Unified end-to-end singing voice synthesis with cross-modality information matching, in *Proc. 31st ACM Int. Conf. Multimedia*, Ottawa, Canada, 2023, pp. 7569–7579.
- [43] J. S. Hwang, S. H. Lee, and S. W. Lee, HiddenSinger: High-quality singing voice synthesis via neural audio codec and latent diffusion models, *Neural Netw.*, vol. 181, p. 106762, 2025.
- [44] K. Chen, Y. Wu, H. Liu, M. Nezhurina, T. Berg-Kirkpatrick, and S. Dubnov, MusicLDM: Enhancing novelty in text-to-music generation using beat-synchronous mixup strategies, in *Proc. ICASSP 2024—2024 IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, Seoul, Republic of Korea, 2024, pp. 1206–1210.
- [45] Q. Huang, D. S. Park, T. Wang, T. I. Denk, A. Ly, N. Chen, Z. Zhang, Z. Zhang, J. Yu, C. Frank, et al., Noise2Music: Text-conditioned music generation with diffusion models, arXiv preprint arXiv: 2302.03917, 2023.
- [46] M. W. Y. Lam, Q. Tian, T. Li, Z. Yin, S. Feng, M. Tu, Y. Ji, R. Xia, M. Ma, X. Song, et al., Efficient neural music generation, arXiv preprint arXiv: 2305.15719, 2023.
- [47] H. Liu, Y. Yuan, X. Liu, X. Mei, Q. Kong, Q. Tian, Y. Wang, W. Wang, Y. Wang, and M. D. Plumbley, AudioLDM 2: Learning holistic audio generation with self-supervised pretraining, *IEEE/ACM Trans. Audio Speech Lang. Process.*, vol. 32, pp. 2871–2883, 2024.
- [48] S. Lei, Y. Zhou, B. Tang, M. W. Y. Lam, H. Liu, J. Wu, S. Kang, Z. Wu, and H. Meng, SongCreator: Lyrics-based universal song generation, arXiv preprint arXiv: 2409.06029, 2024.
- [49] OpenAI, J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, et al., GPT-4 technical report, arXiv preprint arXiv: 2303.08774, 2023.
- [50] N. Bryan-Kinns, B. Banar, C. Ford, C. N. Reed, Y. Zhang, S. Colton, and J. Armitage, Exploring XAI for the arts: Explaining latent space in generative music, arXiv preprint arXiv: 2308.05496, 2023.
- [51] G. Zhao, H. Fu, R. Song, T. Sakai, Z. Chen, X. Xie, and X. Qian, Personalized reason generation for explainable song recommendation, *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 4, pp. 1–21, 2019.
- [52] S. Ding, Z. Liu, X. W. Dong, P. Zhang, R. Qian, C. He, D. Lin, and J. Wang, SongComposer: A large language model for lyric and melody composition in song generation, arXiv preprint arXiv: 2402.17645, 2024.
- [53] H. Kim and K. Choi, LyCon: Lyrics reconstruction from the bag-of-words using large language models, arXiv preprint arXiv: 2408.14750, 2024.
- [54] H. H. Liu and Y. W. Liu, Agent-driven large language models for mandarin lyric generation, in *Proc. 27th Conf. Oriental COCODSA Int. Committee for the Co-Ordination and Standardisation of Speech Databases and Assessment Techniques (O-COCOSDA)*, Hsinchu City, China, 2024, pp. 1–6.
- [55] X. Ma, Y. Wang, M. Y. Kan, and W. S. Lee, AI-lyricist: Generating music and vocabulary constrained lyrics, in *Proc. 29th ACM Int. Conf. Multimedia*, Virtual Event, 2021, pp. 1002–1011.
- [56] Z. Yan, C. Yang, Q. Liang, and X. A. Chen, XCreation: A graph-based crossmodal generative creativity support tool, in *Proc. 36th Annual ACM Symp. on User Interface Software and Technology*, San Francisco, CA, USA, 2023, pp. 1–15.
- [57] J. Aineslahti, AI’s impact on intermediate music production—Mixing and mastering, <https://urn.fi/URN:NBN:fi:amk-2023060621753>, 2023.
- [58] T. Birchneil, Listening without ears: Artificial intelligence in audio mastering, *Big Data Soc.*, vol. 5, no. 2, p. 2053951718808553, 2018.
- [59] J. Sterne and E. Razlogova, Machine learning in context, or learning from LANDR: Artificial intelligence and the platformization of music mastering, *Social Media + Society.*, vol. 5, no. 2, p. 2056305119847525, 2019.
- [60] S. Canyakan, Machine learning in audio mastering: A comparative study, *Journal for the Interdisciplinary Art and Education*, vol. 6, no. 1, pp. 47–65, 2025.
- [61] J. Smit and B. K. Lee, Automatic music mastering using deep learning, in *Proc. Int. Conf. Computational Science and Computational Intelligence (CSCI)*, Las Vegas, NV, USA, 2022, pp. 385–390.
- [62] S. S. Vanka, M. Safi, J. B. Rolland, and G. Fazekas, Adoption of AI technology in the music mixing workflow: An investigation, arXiv preprint arXiv: 2304.03407, 2023.
- [63] B. Aldous, Percussive audio mixing with Wave-U-Nets, <https://www.scribd.com/document/783332117>, 2022.
- [64] X. Liu, Z. Zhu, H. Liu, Y. Yuan, Q. Huang, M. Cui, J. Liang, Y. Cao, Q. Kong, M. D. Plumbley et al., WavJourney: Compositional audio creation with large language models, *IEEE Trans. Audio Speech Lang. Process.*, vol. 33, pp. 2830–2844, 2025.
- [65] J. Liang, H. Zhang, H. Liu, Y. Cao, Q. Kong, X. Liu, W. Wang, M. D. Plumbley, H. Phan, and E. Benetos, Wavcraft: Audio editing and generation with large language models, arXiv preprint arXiv: 2403.09527, 2024.
- [66] T. Karras, S. Laine, M. Aittala, J. Hellsten, J. Lehtinen, and T. Aila, Analyzing and improving the image quality of StyleGAN, in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Seattle, WA, USA, 2020, pp. 8110–8119.
- [67] Y. Shen, J. Gu, X. Tang, and B. Zhou, Interpreting the latent space of GANs for semantic face editing, in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Seattle, WA, USA, 2020, pp. 9243–9252.
- [68] A. van den Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, and K. Kavukcuoglu, WaveNet: A generative model for raw audio, arXiv preprint arXiv: 1609.03499, 2016.
- [69] Z. Borsos, R. Marinier, D. Vincent, E. Kharitonov, O. Pietquin, M. Sharifi, D. Roblek, O. Teboul, D. Grangier, M. Tagliasacchi, et al., AudioLM: A language modeling approach to audio generation, *IEEE/ACM Trans. Audio Speech Lang. Process.*, vol. 31, pp. 2523–2533, 2023.
- [70] D. Yang, J. Yu, H. Wang, W. Wang, C. Weng, Y. Zou, and D. Yu, DiffSound: Discrete diffusion model for text-to-sound generation, arXiv preprint arXiv: 2207.09983, 2022.
- [71] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, arXiv preprint arXiv: 2103.00020, 2021.
- [72] A. Guzhov, F. Raue, J. Hees, and A. Dengel, Audioclip: Extending clip to image, text and audio, in *Proc. ICASSP 2022—2022 IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, Singapore, 2022, pp. 976–980.
- [73] H. Hassinen, Audio and text conditioned abstract sound synthesis through human-AI interaction, <https://aaltodoc.aalto.fi/items/>

- eef8d874-1316-4ae2-b105-61160478d205, 2023.
- [74] T. Zhao, M. Kong, T. Liang, Q. Zhu, K. Kuang, and F. Wu, CLAP: Contrastive language-audio pre-training model for multi-modal sentiment analysis, in *Proc. 2023 ACM Int. Conf. Multimedia Retrieval*, Thessaloniki, Greece, 2023, pp. 622–626.
 - [75] P. Kamath, F. Morreale, P. L. Bagaskara, Y. Wei, and S. Nanayakkara, Sound designer-generative AI interactions: Towards designing creative support tools for professional sound designers, in *Proc. CHI Conf. Human Factors in Computing Systems*, Honolulu, HI, USA, 2024, pp. 1–17.
 - [76] N. M. Norowi, An artificial intelligence approach to concatenative sound synthesis, <https://pearl.plymouth.ac.uk/foahb-theses-other/159/>, 2013.
 - [77] K. Choi, J. Im, L. Heller, B. McFee, K. Imoto, Y. Okamoto, M. Lagrange, and S. Takamichi, Foley sound synthesis at the DCASE 2023 challenge, arXiv preprint arXiv: 2304.12521, 2023.
 - [78] D. Schwarz, G. Beller, B. Verbrugge, and S. Britton, Real-time corpus-based concatenative synthesis with CataRT, in *Proc. 9th International Conference on Digital Audio Effects (DAFx)*, Montreal, Canada, 2006, pp. 279–282.
 - [79] D. Schwarz, Concatenative sound synthesis: The early years, *Journal of New Music Research*, vol. 35, no. 1, pp. 3–22, 2006.
 - [80] N. Bryan-Kinns, B. Banar, C. Ford, C. N. Reed, Y. Zhang, and J. Armitage, Explainable AI and music, in *Artificial Intelligence for Art Creation and Understanding*, L. Mou, ed. Boca Raton, FL, USA: CRC Press, 2024, pp. 1–29.
 - [81] P. W. L. Suckrow, C. J. Weber, and S. Rothe, Diffusion-based sound synthesis in music production, in *Proc. 12th ACM SIGPLAN Int. Workshop on Functional Art, Music, Modelling, and Design*, Milan, Italy, 2024, pp. 55–64.
 - [82] M. Clemens and A. Marasović, MixAssist: An audio-language dataset for co-creative ai assistance in music mixing, arXiv preprint arXiv: 2507.06329, 2025.
 - [83] A. Saffioti, P. Fogel, P. Knudsen, L. de Miranda, and O. Thörn, On human-AI collaboration in artistic performance, in *Proc. 1st International Workshop on New Foundations for Human-Centered AI (NeHuAI) Co-Located with 24th European Conference on Artificial Intelligence (ECAI 2020)*, Santiago de Compostella, Spain, 2020, pp. 38–43.
 - [84] M. Fiorini, Being the artificial player: Good practices in collective human-machine music improvisation, in *ArtsIT, Interactivity and Game Creation*, A. L. Brooks, D. Banakou, and S. Ceperkovic, eds. Cham, Switzerland: Springer, 2025, pp. 310–330.
 - [85] J. B. Smith, Human-AI partnerships in gesture-controlled interactive music systems, <https://repository.gatech.edu/server/api/core/bitstreams/4c81ae3d-1d90-466c-aebd-6c002c5e12ad/content>, 2024.
 - [86] S. Chakraborty, S. Dutta, and J. Timoney, The Cyborg Philharmonic: Synchronizing interactive musical performances between humans and machines, *Humanit. Soc. Sci. Commun.*, vol. 8, no. 1, p. 74, 2021.
 - [87] J. McCormack, P. Hutchings, T. Gifford, M. Yee-King, M. T. Llano, and M. D'inverno, Design considerations for real-time collaboration with creative artificial intelligence, *Org. Sound*, vol. 25, no. 1, pp. 41–52, 2020.
 - [88] G. E. Lewis, Too many notes: Computers, complexity and culture in *Voyager*, *Leonardo Music. J.*, vol. 10, pp. 33–39, 2000.
 - [89] R. Rowe, *Interactive Music Systems: Machine Listening and Composing*. Cambridge, MA, USA: MIT Press, 1993.
 - [90] K. J. Lee, P. Pasquier, and J. Yuri, Revival: Collaborative artistic creation through human-AI interactions in musical creativity, arXiv preprint arXiv: 2503.15498, 2025.
 - [91] G. Assayag, G. Bloch, M. Chemillier, A. Cont, and S. Dubnov, OMax brothers: A dynamic yopology of agents for improvisation learning, in *Proc. 1st ACM Workshop on Audio and Music Computing Multimedia*, Santa Barbara, CA, USA, 2006, pp. 125–132.
 - [92] K. Egon, J. Russell, and R. Julia, AI in art and creativity: Exploring the boundaries of human-machine collaboration, https://osf.io/preprints/osf/g4nd5_v1, 2023.
 - [93] J. Pressing, Cognitive processes in improvisation, *Advances in Psychology*, vol. 19, pp. 345–363, 1984.
 - [94] A. Narayanan and T. Menneer, Quantum artificial neural network architectures and components, *Inf. Sci.*, vol. 128, nos. 3&4, pp. 231–255, 2000.
 - [95] D. Long, L. Liu, S. Gujranian, C. Naomi, and B. Magerko, Visualizing improvisation in LuminAI, an AI partner for co-creative dance, in *Proc. 7th Int. Conf. Movement and Computing*, Virtual Event, 2020, pp. 1–2.
 - [96] Y. Fu, M. Newman, L. Goings, Q. Feng, and J. H. Lee, Exploring the collaborative co-creation process with AI: A case study in novice music production, arXiv preprint arXiv: 2501.15276, 2025.
 - [97] C. Wang, S. Hasler, D. Tanneberg, F. Ocker, F. Joubin, A. Ceravola, J. Deigmoeller, and M. Gienger, LaMI: Large language models for multi-modal human-robot interaction, in *Proc. Extended Abstracts of the CHI Conf. Human Factors in Computing Systems*, Honolulu, HI, USA, 2024, pp. 1–10.
 - [98] M. Zhao, S. Jain, and S. Song, RoCo: Dialectic multi-robot collaboration with large language models, in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, Yokohama, Japan, 2024, pp. 286–299.
 - [99] A. Lim, T. Ogata, and H. G. Okuno, Towards expressive musical robots: A cross-modal framework for emotional gesture, voice and music, *EURASIP J. Audio Speech Music. Process.*, vol. 2012, no. 1, p. 3, 2012.
 - [100] L. Wang, Multimodal robotic music performance art based on GRU-GoogLeNet model fusing audiovisual perception, *Front. Neurobot.*, vol. 17, p. 1324831, 2024.
 - [101] A. De Filippo and M. Milano, Large language models for human-AI co-creation of robotic dance performances, in *Proc. International Joint Conferences on Artificial Intelligence*, Busan, Republic of Korea, 2024, pp. 7627–7635.
 - [102] F. Xu, Y. Xia, and X. Wu, An adaptive control framework based multi-modal information-driven dance composition model for musical robots, *Front. Neurobot.*, vol. 17, p. 1270652, 2023.
 - [103] S. Lu and P. Wang, Multi-dimensional fusion: Transformer and GANs-based multimodal audiovisual perception robot for musical performance art, *Front. Neurobot.*, vol. 17, p. 1281944, 2023.
 - [104] H. Su, W. Qi, J. Chen, C. Yang, J. Sandoval, and M. A. Laribi, Recent advancements in multimodal human-robot interaction, *Front. Neurobot.*, vol. 17, p. 1084000, 2023.
 - [105] L. Scimeca, C. Ng, and F. Iida, Gaussian process inference modelling of dynamic robot control for expressive piano playing, *PLoS One*, vol. 15, no. 8, p. e0237826, 2020.
 - [106] M. Ur Rehman, F. Ahmed, M. Attique Khan, U. Tariq, F. Abdulaziz Alfouzan, N. M. Alzahrani, and J. Ahmad, Dynamic hand gesture recognition using 3D-CNN and LSTM networks, *Comput. Mater. Continua*, vol. 70, no. 3, pp. 4675–4690, 2022.
 - [107] V. S. Yakovyna and V. V. Korniienko, Music emotion classification using a hybrid CNN-LSTM model, *Appl. Aspects Inf. Technol.*, vol. 6, no. 4, pp. 418–430, 2023.
 - [108] P. Hullur, D. Kulkarni, V. Jainapur, S. Chikkamath, S. R. Nirmala, and S. V. Budihal, Music versus speech classification using 3D CNN, in *Proc. International Conference on Recent Trends in Machine Learning, IoT, Smart Cities and Applications*, Singapore: Springer, 2025, pp. 273–282.
 - [109] M. Zeng, X. Tan, R. Wang, Z. Ju, T. Qin, and T. Liu, MusicBERT: Symbolic music understanding with large-scale pre-training, arXiv preprint arXiv: 2106.05630, 2021.
 - [110] C. Liu, X. Zhou, Y. Wu, R. Yang, Z. Wang, L. Zhai, Z. Jia, and Y. Liu, Graph neural networks in EEG-based emotion recognition: A survey, arXiv preprint arXiv: 2402.01138, 2024.
 - [111] X. Hao, H. Li, and Y. Wen, Real-time music emotion recognition based on multimodal fusion, *Alex. Eng. J.*, vol. 116, pp. 586–600, 2025.
 - [112] J. S. G. Cañón, E. Cano, P. Herrera, and E. Gómez, Transfer

- learning from speech to music: Towards language-sensitive emotion recognition models, in *Proc. 28th European Signal Processing Conf. (EUSIPCO)*, Amsterdam, the Netherlands, 2021, pp. 136–140.
- [113] X. Xu, P. Zhang, M. Yan, J. Zhang, and M. Wu, Enhancing zero-shot audio classification using sound attribute knowledge from large language models, arXiv preprint arXiv: 2407.14355, 2024.
- [114] X. Jiang, Y. Zhang, G. Lin, and L. Yu, Music emotion recognition based on deep learning: A review, *IEEE Access*, vol. 12, pp. 157716–157745, 2024.
- [115] Y. Jiang and M. Zheng, The inductive effect of musical mode types on emotions and physiological responses: Chinese pentatonic scales versus Western major/minor modes, *Front. Psychol.*, vol. 15, p. 1414014, 2024.
- [116] S. Luitel, Y. Liu, and M. Anwar, Investigating fairness in machine learning-based audio sentiment analysis, *AI Ethics*, vol. 5, no. 2, pp. 1099–1108, 2025.
- [117] A. Niyazov, E. Mikhailova, and O. Egorova, Content-based music recommendation system, in *Proc. 29th Conf. Open Innovations Association (FRUCT)*, Tampere, Finland, 2021, pp. 274–279.
- [118] P. Faye, S. Sonar, S. Shahare, A. Rakshata, and V. Chafle, Emotion based music recommendation system, *International Journal on Advanced Electrical and Computer Engineering*, vol. 14, no. 1, pp. 186–191, 2025.
- [119] X. Wang and E. Wong, Mapping the emotion-scape in Chinese-language pop song lyrics, 1967–2023: Combining LLM with lexicon-based sentiment analysis, doi: 10.1007/s42803-025-00103-x.
- [120] Y. Huo, M. Jin, and S. You, LSTM-based framework for the synthesis of original soundtracks, *IEEE Access*, vol. 12, pp. 33832–33842, 2024.
- [121] R. Mayya, V. Venkataraman, P. R. Anwesh, and N. Darapaneni, A novel bi-LSTM and transformer architecture for generating tabla music, arXiv preprint arXiv: 2404.05765, 2024.
- [122] L. Qiu, S. Li, and Y. Sung, 3D-DCDAE: Unsupervised music latent representations learning method based on a deep 3D convolutional denoising autoencoder for music genre classification, *Mathematics*, vol. 9, no. 18, p. 2274, 2021.
- [123] J. I. Custal, Analyzing music improvisations using unsupervised machine learning: Towards automatically discovering creative cognition principles, Master dissertation, School of Electrical Engineering and Computer Science, Universitat Politècnica de Catalunya, Barcelona, Spain, 2024.
- [124] T. Kitahara, Music generation using Bayesian networks, in *Machine Learning and Knowledge Discovery in Databases*, Y. Altun, K. Das, T. Mielikäinen, D. Malerba, J. Stefanowski, J. Read, M. Žitnik, M. Ceci, and S. Džeroski, eds. Cham: Springer International Publishing, 2017, pp. 368–372.
- [125] K. Highnam, D. Puzio, S. Luo, and N. R. Jennings, Real-time detection of dictionary DGA network traffic using deep learning, *SN Comput. Sci.*, vol. 2, no. 2, p. 110, 2021.
- [126] J. Klein, B. Pingault, M. Florian, M. C. Heißenbüttel, A. Steinhoff, Z. Song, K. Torres, F. Dirnberger, J. B. Curtis, M. Weile, et al., The bulk van der Waals layered magnet CrSBr is a quasi-1D material, *ACS Nano*, vol. 17, no. 6, pp. 5316–5328, 2023.
- [127] M. Rumiantcev, Transformer-based multimodal framework for music similarity analysis and recommendation systems, in *Proc. 37th Conf. Open Innovations Association (FRUCT)*, Narvik, Norway, 2025, pp. 260–270.
- [128] H. Su, G. Song, K. Huang, J. Wang, and M. Yang, Cross-modal attention alignment network with auxiliary text description for zero-shot sketch-based image retrieval, in *Artificial Neural Networks and Machine Learning—ICANN 2024*, M. Wand, K. Malinová, J. Schmidhuber, and I. V. Tetko, eds. Cham, Switzerland: Springer, 2024, pp. 52–65.
- [129] E. Aras, Style learning and musical mimicry in artificial intelligence: Modern approaches, *Journal of AI, Humanities and New Ethics*, vol. 1, no. 1, pp. 19–32, 2024.
- [130] J. Yang, P. Ren, and X. Kong, Handwriting text recognition based on faster R-CNN, in *Proc. Chinese Automation Congress (CAC)*, Hangzhou, China, 2019, pp. 2450–2454.
- [131] A. Kruspe, Harmonic reasoning in large language models, arXiv preprint arXiv: 2409.05521, 2024.
- [132] M. Krause, C. Weiß, and M. Müller, A cross-version approach to audio representation learning for orchestral music, in *Proc. International Society for Music Information Retrieval (ISMIR) Conference*, San Francisco, CA, USA, 2023, pp. 832–839.
- [133] B. Sarwar, G. Karypis, J. Konstan, and J. Riedl, Item-based collaborative filtering recommendation algorithms, in *Proc. 10th Int. Conf. World Wide Web*, Hong Kong, China, 2001, pp. 285–295.
- [134] Z. Hyung and K. Lee, Recommending music based on probabilistic latent semantic analysis on Korean radio episodes, in *Proc. 9th Int. Conf. Intelligent Information Hiding and Multimedia Signal Processing*, Beijing, China, 2013, pp. 472–476.
- [135] N. Van Dat, P. Van Toan, and T. M. Thanh, Solving distribution problems in content-based recommendation system with Gaussian mixture model, *Appl. Intell.*, vol. 52, no. 2, pp. 1602–1614, 2022.
- [136] A. van den Oord, S. Dieleman, and B. Schrauwen, Deep content-based music recommendation, in *Proc. 27th International Conference on Neural Information Processing Systems—Volume 2*, Lake Tahoe, NV, USA, 2013, pp. 2643–2651.
- [137] B. Hidasi, A. Karatzoglou, L. Baltrunas, and D. Tikk, Session-based recommendations with recurrent neural networks, arXiv preprint arXiv: 1511.06939, 2015.
- [138] X. He and T. S. Chua, Neural factorization machines for sparse predictive analytics, in *Proc. 40th Int. ACM SIGIR Conf. Research and Development in Information Retrieval*, Tokyo, Japan, 2017, pp. 355–364.
- [139] P. Seshadri and P. Knees, Leveraging negative signals with self-attention for sequential music recommendation, arXiv preprint arXiv: 2309.11623, 2023.
- [140] L. Pretet, G. Richard, and G. Peeters, Cross-modal music-video recommendation: A study of design choices, in *Proc. Int. Joint Conf. Neural Networks (IJCNN)*, Shenzhen, China, 2021, pp. 1–9.
- [141] S. Wang, C. Xu, A. S. Ding, and Z. Tang, A novel emotion-aware hybrid music recommendation method using deep neural network, *Electronics*, vol. 10, no. 15, p. 1769, 2021.
- [142] J. Liao, W. Zhou, F. Luo, J. Wen, M. Gao, X. Li, and J. Zeng, SocialLGN: Light graph convolution network for social recommendation, *Inf. Sci.*, vol. 589, pp. 595–607, 2022.
- [143] H. Ding, Y. Ma, A. Deoras, Y. Wang, and H. Wang, Zero-shot recommender systems, arXiv preprint arXiv: 2105.08318, 2021.
- [144] Z. Lu, E. Zhong, L. Zhao, E. W. Xiang, W. Pan, and Q. Yang, Selective transfer learning for cross domain recommendation, in *Proc. 2013 SIAM Int. Conf. Data Mining*, Austin, TX, USA, 2013, pp. 641–649.
- [145] F. Tomasi, J. Cauteruccio, S. Kanoria, K. Ciosek, M. Rinaldi, and Z. Dai, Automatic music playlist generation via simulation-based reinforcement learning, in *Proc. 29th ACM SIGKDD Conf. Knowledge Discovery and Data Mining*, Long Beach, CA, USA, 2023, pp. 4948–4957.
- [146] Y. Deng, X. Qiu, X. Tan, W. Chu, and Y. Xu, LMGT: Optimizing exploration-exploitation balance in reinforcement learning through language model guided trade-offs, arXiv preprint arXiv: 2409.04744v1, 2024.
- [147] S. Doh, K. Choi, and J. Nam, TALKPLAY: Multimodal music recommendation with large language models, arXiv preprint arXiv: 2502.13713, 2025.
- [148] R. Liu and X. Hu, A multimodal music recommendation system with listeners' personality and physiological signals, in *Proc. ACM/IEEE Joint Conf. Digital Libraries in 2020*, Virtual Event, 2020, pp. 357–360.
- [149] X. He, L. Liao, H. Zhang, L. Nie, X. Hu, and T. S. Chua, Neural collaborative filtering, in *Proc. 26th Int. Conf. World Wide Web*, Perth, 2017, pp. 173–182.
- [150] H. Guo, R. Tang, Y. Ye, Z. Li, and X. He, DeepFM: A

- factorization-machine based neural network for CTR prediction, arXiv preprint arXiv: 1703.04247, 2017.
- [151] S. Bai, J. Z. Kolter, and V. Koltun, An empirical evaluation of generic convolutional and recurrent networks for sequence modeling, arXiv preprint arXiv: 1803.01271, 2018.
- [152] M. Li, R. Westerholt, H. Fan, and A. Zipf, Assessing spatiotemporal predictability of LBSN: A case study of three foursquare datasets, *GeoInformatica*, vol. 22, no. 3, pp. 541–561, 2018.
- [153] A. Henry, V. Wiratama, A. Afilipoaie, H. Ranaivoson, and E. Arrivé, Impacts of AI on music consumption and fairness, *Emerg. Medium.*, vol. 2, no. 3, pp. 382–396, 2024.
- [154] F. Wang and A. Slowik, AI-empowered consumer behavior analysis for trustworthy track recommendation over musical dance electronic products, *Heliyon*, vol. 10, no. 18, p. e37633, 2024.
- [155] S. Olayeni, The impact of artificial intelligence (AI) in music business industry, <https://urn.fi/URN:NBN:fi:amk-2023102327854>, 2023.
- [156] I. M. Enholm, E. Papagiannidis, P. Mikalef, and J. Krogstie, Artificial intelligence and business value: A literature review, *Inf. Syst. Front.*, vol. 24, no. 5, pp. 1709–1734, 2022.
- [157] G. Hadjeres, F. Pachet, and F. Nielsen, DeepBach: A steerable model for bach chorales generation, arXiv preprint arXiv: 1612.01010, 2017.
- [158] M. Civit, J. Civit-Masot, F. Cuadrado, and M. J. Escalona, A systematic review of artificial intelligence-based music generation: Scope, applications, and future trends, *Expert Syst. Appl.*, vol. 209, p. 118190, 2022.
- [159] D. Bryce, Artificial intelligence and music: Analysis of music generation techniques via deep learning and the implications of AI in the music industry, https://digitalcommons.bryant.edu/honors_data_science/12/, 2024.
- [160] J. Alem, The impact of artificial intelligence on the music industry, <http://hdl.handle.net/10400.14/46557>, 2024.
- [161] R. S. Huang, B. L. T. Sturm, and A. Holzapfel, De-centering the west: East Asian philosophies and the ethics of applying artificial intelligence to music, in *Proc. 22nd International Society for Music Information Retrieval (ISMIR) Conference*, Virtual Event, 2021, pp. 301–309.
- [162] G. Born, J. Morris, F. Diaz, and A. Anderson, Artificial intelligence, music recommendation, and the curation of culture, <http://hdl.handle.net/1807/129105>, 2021.
- [163] B. L. T. Sturm, M. Iglesias, O. Ben-Tal, M. Miron, and E. Gómez, Artificial intelligence and music: Open questions of copyright law and engineering praxis, *Arts*, vol. 8, no. 3, p. 115, 2019.
- [164] M. V. B. Pineda, New entrepreneurial strategies using the new technologies of artificial intelligence and NFT in the field of the music industry, *J. Digit. Econ.*, vol. 3, pp. 260–274, 2024.
- [165] E. Drott, Copyright, compensation, and commons in the music AI industry, *Creat. Ind. J.*, vol. 14, no. 2, pp. 190–207, 2021.
- [166] A. M. Horzyk, How AI affects our understanding of musical works that should be protected by copyright, in *Proc. Int. Joint Conf. Neural Networks (IJCNN)*, Gold Coast, Australia, 2023, pp. 1–8.
- [167] A. Green, OECD employment outlook 2023: Artificial intelligence and the labour market, https://www.oecd.org/en/publications/oecd-employment-outlook-2023_08785bba-en.html, 2023.
- [168] CISAC, Global collections report 2024, <https://www.cisac.org/services/reports-and-research/global-collections-report-2024>, 2024.
- [169] K. Goldhammer, M. Birkel, L. Mackuth, and D. Komma, AI and music: Market development of AI in the music sector and impact on music authors and creators in Germany and France, <https://www.gema.de/documents/d/guest/gema-sacem-goldmedia-ai-and-music-pdf>, 2024.
- [170] This is music 2024, <https://www.ukmusic.org/research-reports/this-is-music-2024/>, 2024.
- [171] A. Kawakami, K. Furukawa, K. Katahira, K. Kamiyama, and K. Okanoya, Relations between musical structures and perceived and felt emotions, *Music. Percept.*, vol. 30, no. 4, pp. 407–417, 2013.
- [172] F. Pachet, P. Roy, and B. Carré, Assisted music creation with flow machines: Towards new categories of new, in *Handbook of Artificial Intelligence for Music*, E. R. Miranda, ed. Cham, Switzerland: Springer International Publishing, 2021, pp. 485–520.
- [173] European Union, Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act) (Text with EEA relevance), <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>, 2024.
- [174] U. S. Copyright Office, Copyright and Artificial Intelligence, Part 2: Copyrightability, <https://www.copyright.gov/ai/Copyright-and-Artificial-Intelligence-Part-2-Copyrightability-Report.pdf>, 2024.
- [175] Copyright and artificial intelligence, <https://www.gov.uk/government/consultations/copyright-and-artificial-intelligence/copyright-and-artificial-intelligence>, 2024.
- [176] General understanding on AI and copyright in Japan, https://www.bunka.go.jp/english/policy/copyright/pdf/94055801_01.pdf, 2024.
- [177] Spotify removes ‘tens of thousands’ of AI-generated songs: Here’s why, <https://www.forbes.com/sites/ariannajohnson/2023/05/09/spotify-removes-tens-of-thousands-of-ai-generated-songs-heres-why/>, 2023.
- [178] Provisions on the management of deep synthesis of Internet information services, (in Chinese), https://www.cac.gov.cn/2022-12/11/c_1672221949354811.htm, 2024.
- [179] Copyright Act 2021, Singapore, <https://sso.agc.gov.sg/Act/CA202>, 2024.
- [180] X. Han, Innovation of teaching mechanism of music course integrating artificial intelligence technology: ITMMCAI-MCA-ACNN approach, *Egypt. Inform. J.*, vol. 29, p. 100608, 2025.
- [181] L. Zhang, Fusion artificial intelligence technology in music education teaching, *J. Electr. Syst.*, vol. 19, no. 4, pp. 178–195, 2024.
- [182] I. Celik, M. Dindar, H. Muukkonen, and S. Järvelä, The promises and challenges of artificial intelligence for teachers: A systematic review of research, *TechTrends*, vol. 66, no. 4, pp. 616–630, 2022.
- [183] X. Han, Design of vocal music education system based on VR technology, *Procedia Comput. Sci.*, vol. 208, pp. 5–11, 2022.
- [184] N. V. Ashykhmina, The use of ICT tools in training future music teachers to conduct distance learning for primary school students, <http://dspace.pdpu.edu.ua/handle/123456789/22206>, 2025.
- [185] I. Dobrača, The application of the GCompris software as an educational tool in musical culture teaching, <https://dialnet.unirioja.es/servlet/articulo?codigo=10088448>, 2025.
- [186] E. Patiño, M. López, S. Lantarón, and J. Rodrigo, Crochet as a work tool in university mathematics studies, <https://dialnet.unirioja.es/servlet/articulo?codigo=10088449>, 2025.
- [187] M. J. P. de Felipe, M. P. R. Sainz, and B. M. García-Villalba, Exploring microbiology and infectious disease through the art: Enferm_Arte Project, <https://investigacionusp.ceu.es/en/ipublic/item/10219919>, 2025.
- [188] S. Y. V. Maraño, Discovering sound art in the classroom: An interdisciplinary workshop, <http://hdl.handle.net/10045/153281>, 2025.
- [189] A. J. Esparcia, M. A. P. Fernández, M. T. T. Reynolds, and J. J. G. Arranz, Revaluating the book in the learning of art history: Renewed methodologies around textual and hypertextual media, <https://dialnet.unirioja.es/servlet/articulo?codigo=10088443>, 2025.
- [190] X. Wang, Design of vocal music teaching system platform for music majors based on artificial intelligence, *Wirel. Commun. Mob. Comput.*, vol. 2022, no. 1, p. 5503834, 2022.
- [191] I. F. Váli, Falerone art colony 2024—The whole city is a studio, in *Proc. CIVAE 2025: 7th Interdisciplinary and Virtual Conference on Arts in Education*, Virtual Event, 2025, pp. 42–47.
- [192] C. A. N. Mazlan, S. I. Koning, M. Y. M. Yusoff, D. S. H. A. Hassim, R. Hidayatullah, and S. Jammongsarn, Artificial

- intelligence in music education: Trends, challenges, and future directions, in *Proc. 2025 5th International Conference on Artificial Intelligence and Education (ICAIE)*, Suzhou, China, 2025, pp. 713–717.
- [193] R. S. Camarero and J. O. Revilla, Underlying stereotypes in visual arts approaches, <https://dialnet.unirioja.es/servlet/articulo?codigo=10088434>, 2025.
- [194] S. N. Lalanda, Global competencies and music education: Contributions to an integral learning development, <https://iris.uniroma3.it/handle/11590/507996>, 2025.
- [195] I. Raudonikytė, The aesthetic aspect of sustainability and educational potential of the interiors of a Lithuanian country house: The evolution of the relationship between function, aesthetics and ecology in the 20th-21st centuries, <https://epubl.ktu.edu/object/elaba:227228021/index.html>, 2025.
- [196] A. Raglio, P. Baiardi, G. Vizzari, M. Imbriani, M. Castelli, S. Manzoni, F. Vico, and L. Manzoni, Algorithmic music for therapy: Effectiveness and perspectives, *Appl. Sci.*, vol. 11, no. 19, p. 8833, 2021.
- [197] Q. Ding, Evaluation of the efficacy of artificial neural network-based music therapy for depression, *Comput. Intell. Neurosci.*, vol. 2022, p. 9208607, 2022.
- [198] Z. Wang, X. Guan, E. Li, and B. Dong, A study on music therapy aimed at psychological trauma recovery for bereaved families driven by artificial intelligence, *Front. Psychol.*, vol. 15, p. 1436324, 2024.
- [199] J. S. Rahman, T. Gedeon, S. Caldwell, R. Jones, and Z. Jin, Towards effective music therapy for mental health care using machine learning tools: Human affective reasoning and music genres, *J. Artif. Intell. Soft Comput. Res.*, vol. 11, no. 1, pp. 5–20, 2021.
- [200] A. Tandon, A systematic review of AI-based therapeutic music generation systems, in *Evolution in Computational Intelligence*, V. Bhateja, P. Patel, and J. Tang, eds. Singapore: Springer, 2024, pp. 163–179.
- [201] J. Sun, J. Yang, G. Zhou, Y. Jin, and J. Gong, Understanding human-AI collaboration in music therapy through co-design with therapists, in *Proc. CHI Conf. Human Factors in Computing Systems*, Honolulu, HI, USA, 2024, pp. 1–21.
- [202] R. Bazaitu, A. D. Bazaitu, and T. V. Axinte, Uses of artificial intelligence in the music therapy, *Int. J. Adv. Multidiscip. Res. Stud.*, vol. 4, no. 3, pp. 1595–1598, 2024.